MILITARY SPECIFICATION

MICROCIRCUITS, DIGITAL, 8192-BIT SCHOTTKY, BIPOLAR, PROGRAMMABLE READ-ONLY MEMORY (PROM), MONOLITHIC SILICON

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for monolithic silicon, PROM microcircuits which employ thin film nichrome (NiCr) resistors, zapped vertical emitter, tungsten (W), titanium-tungsten (TiW), or platinum silicide as the fusible link or programming element. Two product assurance classes and a choice of case outlines and lead finishes are provided for each type and are reflected in the part number. A special test requirement is included in this specification to screen against devices which may contain excess moisture in the package materials or internal atmosphere (see freeze-out test of 4.2d).

1.2 Part number. The part number shall be in accordance with MIL-M-38510.

1.2.1 Device types. The device types shall be as follows:

Device type	Circuit	Access times (ns)
01	2048 word/4 bits per word PROM with uncommitted collector	125
02, 08, 10	2048 word/4 bits per word PROM with active pullup and a third high-impedance state output	125, 90, 55
03	1024 word/8 bits per word PROM with uncommitted collector	90
04, 09	1024 word/8 bits per word PROM with active pullup and a third high-impedance state output	90, 55
05	1024 word/8 bits per word PROM with active pullup and a third high-impedance state output	90
06	1024 word/8 bits per word PROM with uncommitted collector	90

1.2.2 Device class. The device class shall be the product assurance level as defined in MIL-M-38510.

1.2.3 Case outlines. The case outlines shall be designated as follows:

Letter	Case outline (see MIL-M-38510, appendix C)
J	D-3 (24-lead, 1/2" x 1 1/4") dual-in-line package
K	F-6 (24-lead, 3/8" x 5/8") flat package
V	D-6 (18-lead, 1/4" x 15/15") dual-in-line package
X	See figure 1 (18-lead, 3/8" x 3/8") flat package

1.3 Absolute maximum ratings.

Supply voltage range	-0.5 V to +7.0 V
Input voltage range	-1.5 V at -10 mA to +5.5 V
Storage temperature range	-65°C to +150°C

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Lead temperature (soldering, 10 seconds)	+300°C
Thermal resistance, junction-to-case (O_{JC}) 1/	2/
Output voltage	-0.5 V to $+V_{CC}$
Output sink current	100 mA
Maximum power dissipation (P_D) 3/:	
Device types 01, 02, 08, and 10	950 mW
Device types 03, 04, 05, 06, and 09	1.1 W
Maximum junction temperature (T_J)	+175°C

1.4 Recommended operating conditions.

Supply voltage	4.5 V minimum to 5.5 V maximum
Minimum high-level input voltage (V_{IH})	2.0 V
Maximum low-level input voltage (V_{IL})	0.8 V
Normalized fanout (each output):	
Device types 01, 02, 08, and 10	12 mA 4/
Device types 03, 04, 05, 06, and 09	8 mA 4/
Case operating temperature range (T_C)	-55°C to +125°C

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification and standard. The following specification and standard form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards and supplement thereto, cited in the solicitation.

SPECIFICATION

MILITARY

MIL-M-38510 - Microcircuits, General Specification for.

STANDARD

MILITARY

MIL-STD-883 - Test Methods and Procedures for Microelectronics.

(Copies of the specification and standard required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Detail specification. The individual item requirements shall be in accordance with MIL-M-38510, and as specified herein. When manufacturer-programmed devices are delivered to the user, an altered item drawing shall be prepared by the contracting activity to specify the required program configuration.

3.2 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-M-38510 and herein.

1/ Heat sinking is recommended to reduce the junction temperature.
2/ Case outlines J and V - 40°C/W maximum
Case outline K - 60°C/W maximum
Case outline X - 35°C/W maximum
3/ Must withstand the added P_D due to short-circuit test (e.g., I_{OS}).
4/ 16 mA for circuits B, D, and F devices.

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3.2.1 Terminal connections. The terminal connections shall be as specified on figure 2.

3.2.2 Truth tables.

3.2.2.1 Unprogrammed devices. The truth tables for unprogrammed devices for contracts involving no altered item drawing shall be as specified on figure 3. When required in group A, B, or C inspection (see 4.4), the devices shall be programmed by the manufacturer prior to test in a checkerboard pattern (a minimum of 50 percent of the total number of bits programmed) or to any altered item drawing pattern which includes at least 25 percent of the total number of bits programmed.

3.2.2.2 Programmed devices. The truth table for programmed devices shall be as specified by the altered item drawing.

3.2.3 Functional block diagrams. The functional block diagrams shall be as specified on figure 4.

3.2.4 Case outlines. The case outlines shall be as specified in 1.2.3.

3.3 Lead material and finish. The lead material and finish shall be in accordance with MIL-M-38510 (see 6.5).

3.4 Electrical performance characteristics. Unless otherwise specified, the electrical performance characteristics of table I apply over the full recommended case operating temperature range.

3.5 Electrical test requirements. The electrical test requirements for each device class shall be as indicated by the subgroups shown in table II and, where applicable, by the altered item drawing. The subgroup tests shall be as specified in in table III.

3.6 Marking. Marking shall be in accordance with MIL-M-38510. For programmed devices, the altered item drawing number shall be added to the marking by the programming activity.

3.7 Processing options. Since the PROM is an unprogrammed memory capable of being programmed by either the manufacturer or the user in a wide variety of configurations, two processing options are provided for selection in the contract, using an altered item drawing.

3.7.1 Unprogrammed PROM delivered to the user. All testing shall be verified through group A testing as defined in 3.2.2.1 and tables II and III. It is recommended that users perform subgroups 7 and 9 after programming to verify the specific program configuration.

3.7.2 Manufacturer-programmed PROM delivered to the user. All testing requirements and quality assurance provisions herein, including the requirements of the altered item drawing, shall be satisfied by the manufacturer prior to delivery.

3.8 Microcircuit group assignment. The devices covered by this specification shall be in microcircuit group number 14 (see MIL-M-38510, appendix E).

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection procedures shall be in accordance with MIL-M-38510 and methods 5005 and 5007, as applicable, of MIL-STD-883, except as modified herein.

4.2 Screening. Screening shall be in accordance with method 5004 of MIL-STD-883, and shall be conducted on all devices prior to qualification and quality conformance inspection. The following additional criteria shall apply:

a. Interim and final electrical tests shall be as specified in table II; the interim electrical tests prior to burn-in are optional at the discretion of the manufacturer.

b. Burn-in test (method 1015 of MIL-STD-883), using the circuits shown on figure 5 or equivalent.

(1) Test condition D or E.

(2) T_A = +125°C minimum.

c. The percent defective allowable (PDA) shall be as specified in MIL-M-38510.

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TABLE I. Electrical performance characteristics.

Test	Symbol	Conditions 1/	Device	Limits		
				Min	Max	
High-level output voltage	V_{OH}	V_{CC} = 4.5 V, I_{OH} = -2 mA V_{IH} = 2.0 V, V_{IL} = 0.8 V	02,04, 05,08, 09,10	2.4		V
Low-level output voltage	V_{OL}	V_{CC} = 4.5 V, I_{OL} = 12 mA 2/ V_{IH} = 2.0 V, V_{IL} = 0.8 V	01,02, 08,10		0.5	V
		V_{CC} = 4.5 V, I_{OL} = 8 mA 2/	03,04, 05,06, 09			
Input clamp voltage	V_{IC}	V_{CC} = 4.5 V, I_{IN} = -10 mA T_C = +25°C	All		-1.5	V
Maximum collector cut-off current	I_{CEX}	V_{CC} = 5.5 V, V_O = 5.2 V	01,03, 06		100	µA
High-impedance (off-state) output high current	I_{OHZ}	V_{CC} = 5.5 V, V_O = 5.2 V	02,04, 05,08, 09,10		100	µA
High-impedance (off-state) output low current	I_{OLZ}	V_{CC} = 5.5 V, V_O = 0.5 V	02,04, 05,08, 09,10		-100	µA
High-level input current	I_{IH1}	V_{CC} = 5.5 V, V_{IN} = 5.5 V	All		50	µA
	I_{IH2}	V_{CC} = 5.5 V, V_{IN} = 4.5 V, special programming pin	03,04, 06,09		100	µA
Low-level input current	I_{IL}	V_{CC} = 5.5 V, V_{IN} = 0.5 V	All	-1.0	-250	µA
Short-circuit output current	I_{OS}	V_{CC} = 5.5 V, V_O = 0.0 V 3/ V_{IH} = 2.0 V, V_{IL} = 0.8 V	02,04, 05,08, 09,10	-15	-100	mA
Supply current	I_{CC}	V_{CC} = 5.5 V, V_{IN} = 0 V, outputs = open	01,02		170	mA
			03,04, 05,06, 08,09, 10		185	

See footnotes at end of table.

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TABLE I. Electrical performance characteristics - Continued.

Test	Symbol	Conditions 1/	Device type	Limits Min	Limits Max	Unit
Propagation delay time, high-to-low level logic, address to output	t_{PHL1}	V_{CC} = 4.5 V and 5.5 V, C_L = 30 pF (see figure 8)	08		90	ns
			01,02		125	
			03,04, 05,06		90	
			09,10		55	
Propagation delay time, low-to-high level logic, address to output	t_{PLH1}		08		90	ns
			01,02		125	
			03,04, 05,06		90	
			09,10		55	
Propagation delay time, high-to-low level logic, enable to output	t_{PHL2}		08		50	ns
			01,02		60	
			03,04, 05,06		50	
			09,10		30	
Propagation delay time, low-to-high level logic, enable to output	t_{PLH2}		08		50	ns
			01,02		60	
			03,04, 05,06		50	
			09,10		30	

1/ Complete terminal conditions shall be as specified in table III.
2/ I_{OL} = 16 mA for circuits B, D, and F devices.
3/ Not more than one output shall be grounded at one time. Output shall be at high logic level prior to test.

TABLE II. Electrical test requirements.

MIL-STD-883 test requirements	Subgroups (see table III) 1/ 2/ 3/	
	Class S devices	Class B devices
Interim electrical parameters (method 5004)	1	1
Final electrical tests (method 5004) for unprogrammed devices	1*,2,3,7*,8	1*,2,3,7*,8
Final electrical tests (method 5004) for programmed devices	1*,2,3,7*, 8,9,10,11	1*,2,3,7*,8,9
Group A electrical tests (method 5005)	1,2,3,7,8, 9,10,11	1,2,3,7,8, 9,10,11
Group B electrical tests (method 5005, subgroup 5)	1,2,3,7,8, 9,10,11	N/A
Group C end-point electrical tests (method 5005)	N/A	1,2,3,7,8
Group D end-point electrical tests (method 5005)	1,2,3,7,8	1,2,3,7,8

1/ * indicates PDA applies to subgroups 1 and 7 (see 4.2c).
2/ Any or all subgroups may be combined when using high-speed testers.
3/ Subgroups 7 and 8 shall consist of verifying the pattern specified.

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d. The freeze-out test shall be conducted as a 100-percent screen on all class S devices having nichrome or platinum silicide as the fusing link. Within no more than 24 hours after completion of burn-in and prior to the final electrical tests, all devices having nichrome or platinum silicide as the fusing link (see 3.7.1 and 3.7.2) shall be subjected to a freeze-out test. If more than 24 hours have elapsed subsequent to the +125°C burn-in exposure, devices shall be conditioned at +125°C for a minimum of 5 hours immediately prior to the freeze-out test. When the freeze-out test is performed, the +25°C final electrical tests shall be completed within 96 hours after the freeze-out test. The freeze-out test shall be conducted as follows:

Step 1. Connect devices in the electrical configuration of figure 7 or in the burn-in configuration of figure 5 with the bias cycled, 3 minutes on and 3 minutes off, throughout the duration of test.

Step 2. Reduce device temperature to T_C = -10°C ±2°C with bias cycled and maintain at that temperature for a minimum of 5 hours.

Step 3. With the cycled bias maintained, allow T_C to go to room temperature (by removal from the cold chamber or termination of forced cooling but with no forced heating) and retain for a minimum of 19 hours subsequent to the completion of the 5-hour cold soak. T_C shall not exceed +35°C during this period.

Step 4. Remove bias and subject all devices to subgroup 1 final electrical tests to establish continuity of the fusible links and remove all failed devices from the lot. Count them as screening rejects subject to the PDA requirements of 4.2c.

e. Class B devices processed to an altered item drawing may be programmed either before or after burn-in at the manufacturer's discretion. The required electrical testing shall include, as a minimum, the final electrical tests for programmed devices as specified in table II herein. Class S devices processed by the manufacturer to an altered item drawing shall be programmed prior to burn-in.

4.3 Qualification inspection. Qualification inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4). Qualification data for subgroups 7 through 11 shall be attributes only.

4.3.1 Qualification extension. When authorized by the qualifying activity, for qualification inspection, if a manufacturer qualifies one device type which is manufactured identically (i.e., same die, process, and package) to other device types on this specification, then the other device types may be part I qualified by conducting only group A electrical tests and submitting data in accordance with MIL-M-38510, appendix D (i.e., groups B, C, and D tests are not required).

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-M-38510, and as specified herein. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4).

4.4.1 Group A inspection. Group A inspection shall be in accordance with table I of method 5005 of MIL-STD-883 and as follows:

a. Electrical test requirements shall be as specified in table II herein.

b. Subgroups 4, 5, and 6 shall be omitted.

c. For unprogrammed devices, a sample shall be selected to satisfy the programmability requirements prior to performing subgroups 9, 10, and 11. Twelve devices shall be submitted to programming (see 3.2.2.1). If more than 2 devices fail to program, the lot shall be rejected. At the manufacturer's option, the sample may be increased to 24 total devices with no more than 4 total device failures allowed.

d. For unprogrammed devices, 10 devices from the programmability sample shall be subjected to the requirements of group A, subgroups 9, 10, and 11. If more than 2 total devices fail in all three subgroups, the lot shall be rejected. At the manufacturer's option, the sample may be increased to 20 total devices with no more than 4 total device failures allowed.

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4.4.2 Group B inspection. Group B inspection shall be in accordance with table II of method 5005 of MIL-STD-883 and as follows:

a. For qualification inspection, at least 50 percent of the sample selected for testing in subgroup 5 shall be programmed (see 3.2.2). For quality conformance inspection, the programmability sample (see 4.4.1c) shall be included in the subgroup 5 tests.

b. Steady-state life tests for class S devices shall be in accordance with subgroup 5, table IIa of method 5005 of MIL-STD-883, using a circuit submitted to the qualifying activity for approval. If the alternate burn-in conditions are used, the circuit on figure 5 or equivalent shall be used.

4.4.3 Group C inspection. Group C inspection shall be in accordance with table III of method 5005 of MIL-STD-883 and as follows:

a. End-point electrical parameters shall be as specified in table II herein.

b. Steady-state life tests (method 1005 of MIL-STD-883, test condition D or E), using the circuits shown on figure 5 or equivalent.

(1) T_A = +125°C minimum.

(2) Test duration = 1,000 hours, except as permitted by appendix B of MIL-M-38510 and method 1005 of MIL-STD-883.

c. For qualification inspection, at least 50 percent of the sample selected for testing in subgroup 1 shall be programmed (see 3.2.2). For quality conformance inspection, the programmability sample (see 4.4.1c) shall be included in the subgroup 1 tests.

4.4.4 Group D inspection. Group D inspection shall be in accordance with table IV of method 5005 of MIL-STD-883. End-point electrical tests shall be as specified in table II herein.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows:

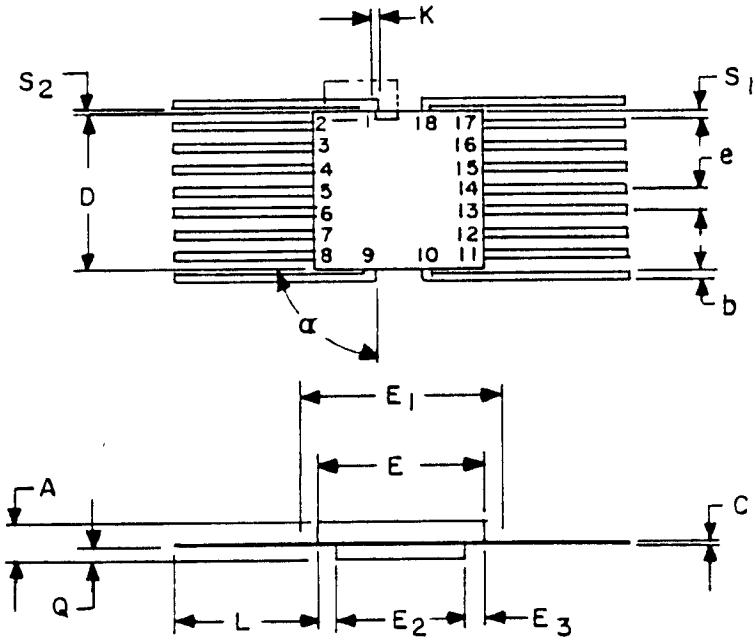
4.5.1 Voltage and current. All voltages given are referenced to the microcircuit ground terminal. Currents given are conventional and positive when flowing into the referenced terminal.

4.6 Programming procedures identification. The programming procedures to be utilized shall be identified by the manufacturer's circuit designator. The circuit designator is cross-referenced in 6.6 herein with the manufacturer's symbol or CAGE number.

Text continues on page 48.

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SYMBOL	INCHES		MILLIMETERS		NOTES
	MIN	MAX	MIN	MAX	
A	.045	.085	1.14	2.16	
b	.015	.020	.38	.51	5
c	.003	.006	.08	.15	5
D	.340	.380	8.64	9.65	
E	.340	.380	8.64	9.65	
E_1		.400		10.37	3
E_2	.260	.290	6.60	7.37	
E_3	.025		.63		
e	.050 BSC		1.27 BSC		4,6
K	.008	.015	.20	.38	9
L	.250	.330	6.35	8.38	
Q	.010	.040	.25	1.02	2
S_1	.005		.13		7,8
S_2	.004		.10		10
α	30°	90°	30°	90°	



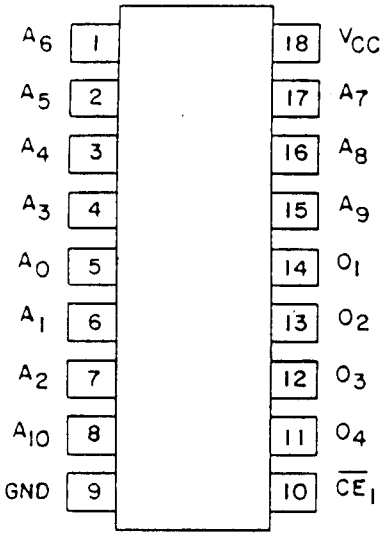
NOTES:
1. Index area: A tab (dim. K) is used to identify pin one.
2. Dimension Q shall be measured at the point of exit of the lead from the body.
3. This dimension allows for off-center lid, meniscus and glass overrun.
4. The basic pin spacing is .050 (1.27 mm) between centerlines. Each pin centerline shall be located within ±.005 (0.13 mm) of its exact longitudinal position relative to pins 1 and 18.
5. All leads-increase may limit by .003 (0.08 mm) measured at the center of the flat, when finish A is applied.
6. Sixteen spaces.
7. Applies to all four corners (leads no. 2, 8, 11, and 17).
8. Dimension S_1 may be .000 (0.00 mm) if leads are brazed to the metallized ceramic body (see appendix C of MIL-M-38510).
9. Optional, see note 1. If pin one identification mark is used in addition to this tab the minimum limit of dimension K does not apply.
10. Applies to leads no. 1, 9, 10, and 18.

FIGURE 1. Case outline X (18-lead, .375 inch X .375 inch).

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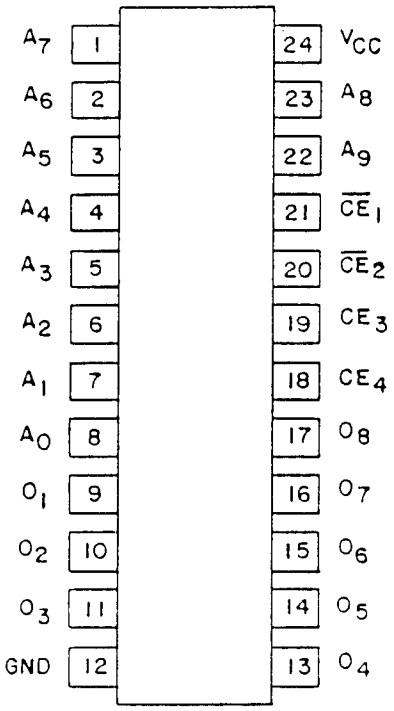
Device types 01, 02, 08 and 10

Case V



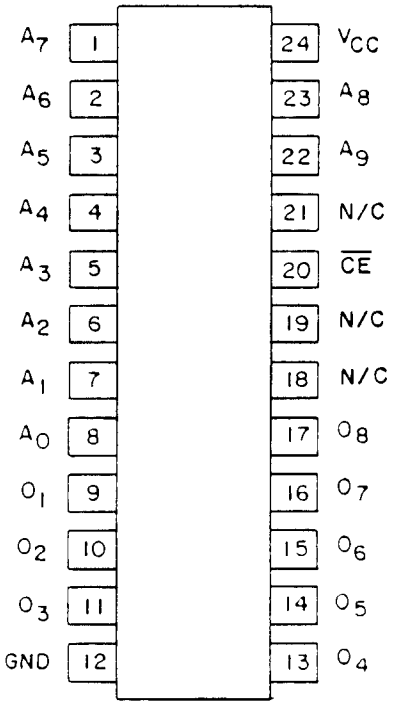
Device types 03, 04 and 09

Cases J and K



Device types 05 and 06

Cases J and K



Device types 01, 02 and 08

Case X

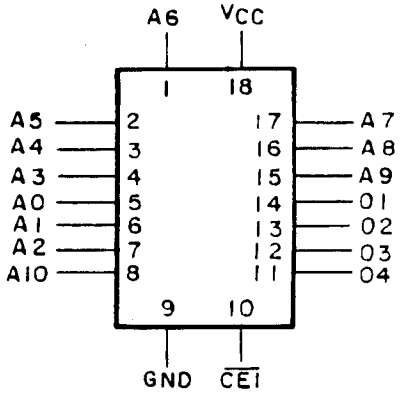


FIGURE 2. Terminal connections.

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Device types 01, 02, 08 and 10 (see notes 1, 2 and 3)

Word no.	Enable	Address											Data			
	$\overline{CE}_1$	A_{10}	A_9	A_8	A_7	A_6	A_5	A_4	A_3	A_2	A_1	A_0	O_1	O_2	O_3	O_4
NA	L	X	X	X	X	X	X	X	X	X	X	X	See note 5			
NA	H	X	X	X	X	X	X	X	X	X	X	X	OC	OC	OC	OC

Device types 03, 04 and 09 (see notes 1, 2, 3 and 4)

Word no.	Enable				Address										Data							
	$\overline{CE}_1$	$\overline{CE}_2$	CE_3	CE_4	A_9	A_8	A_7	A_6	A_5	A_4	A_3	A_2	A_1	A_0	O_1	O_2	O_3	O_4	O_5	O_6	O_7	O_8
NA	L	L	H	H	X	X	X	X	X	X	X	X	X	X	See note 5							
NA	L	H	H	H	X	X	X	X	X	X	X	X	X	X	OC	OC	OC	OC	OC	OC	OC	OC
NA	H	L	H	H	X	X	X	X	X	X	X	X	X	X	OC	OC	OC	OC	OC	OC	OC	OC
NA	H	H	L	H	X	X	X	X	X	X	X	X	X	X	OC	OC	OC	OC	OC	OC	OC	OC
NA	H	H	L	L	X	X	X	X	X	X	X	X	X	X	OC	OC	OC	OC	OC	OC	OC	OC

Device types 05 and 06 (see notes 1, 2 and 3)

Word no.	Enable	Address										Data							
	$\overline{CE}$	A_9	A_8	A_7	A_6	A_5	A_4	A_3	A_2	A_1	A_0	O_1	O_2	O_3	O_4	O_5	O_6	O_7	O_8
NA	L	X	X	X	X	X	X	X	X	X	X	See note 5							
NA	H	X	X	X	X	X	X	X	X	X	X	OC	OC	OC	OC	OC	OC	OC	OC

NOTES:
1. NA = Not applicable.
2. X = Input may be high level, low level, or open circuit.
3. OC = Open circuit (high resistance output).
4. Program readout can only be accomplished with both enable inputs at low level.
5. The outputs for an unprogrammed device shall be high for circuits, A, B (device types 03 and 04), and F, and shall be low for circuits B (device types 01, 02, and 08), C, D, E and G.

FIGURE 3. Truth tables (unprogrammed).

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Device types 01, 02 and 08 (Circuit B)

Device types 01 and 02 (Circuit A)

Device type 02 (Circuit F)

Device types 03 and 04 (Circuit E)

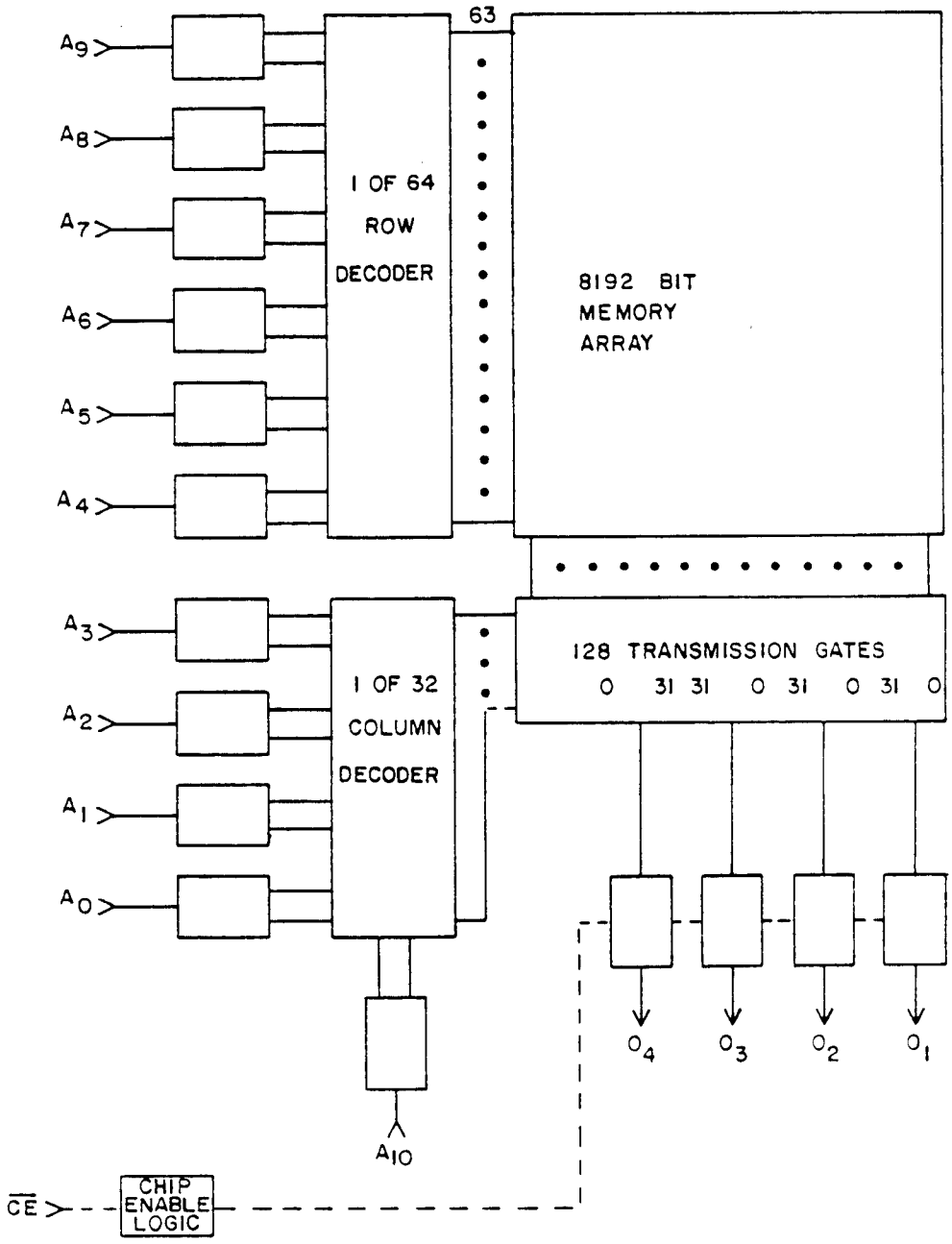


FIGURE 4. Functional block diagrams.

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Device types 01, 02 and 10

Circuit C

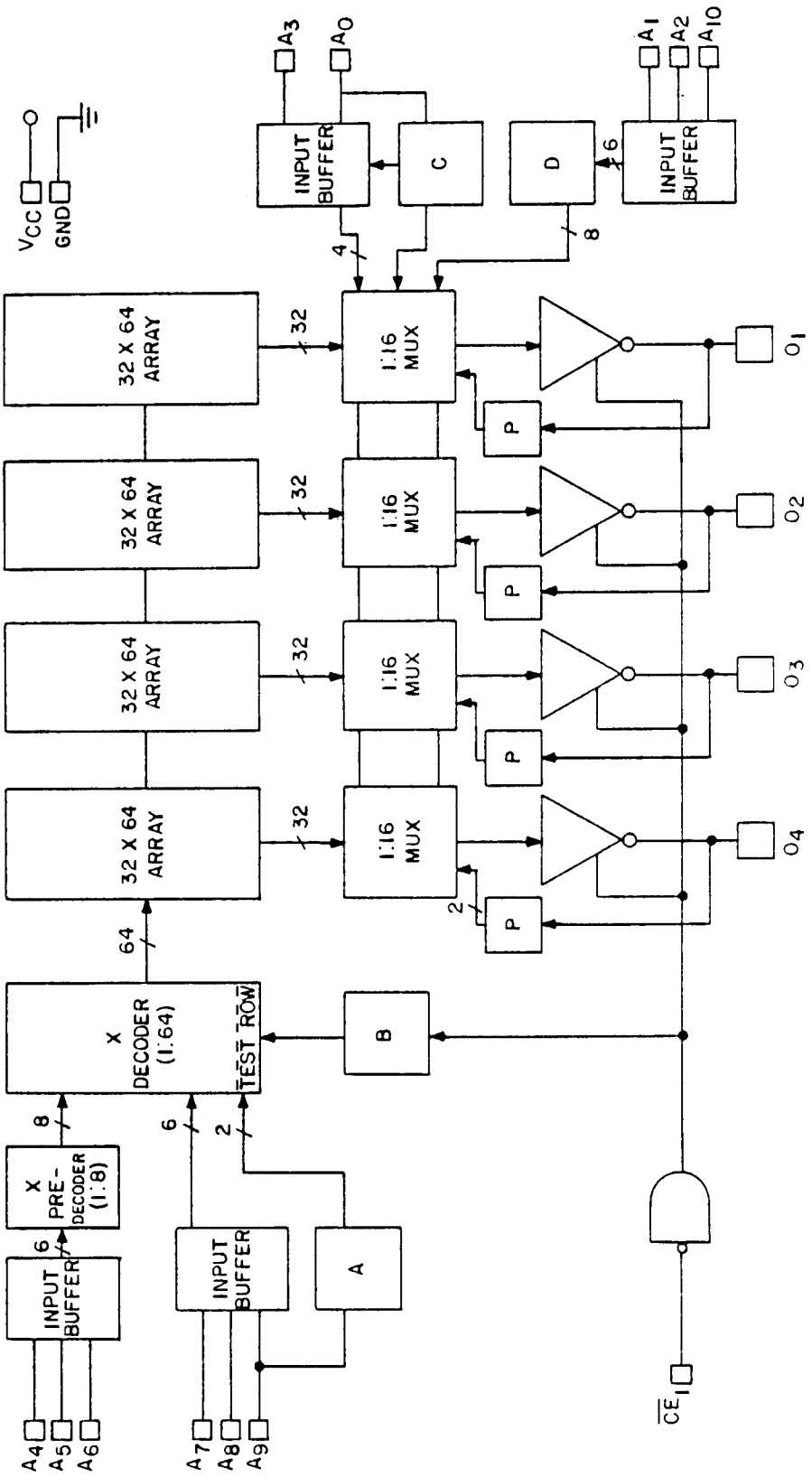


FIGURE 4. Functional block diagrams - Continued.

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Device types 03 and 04

Circuit A

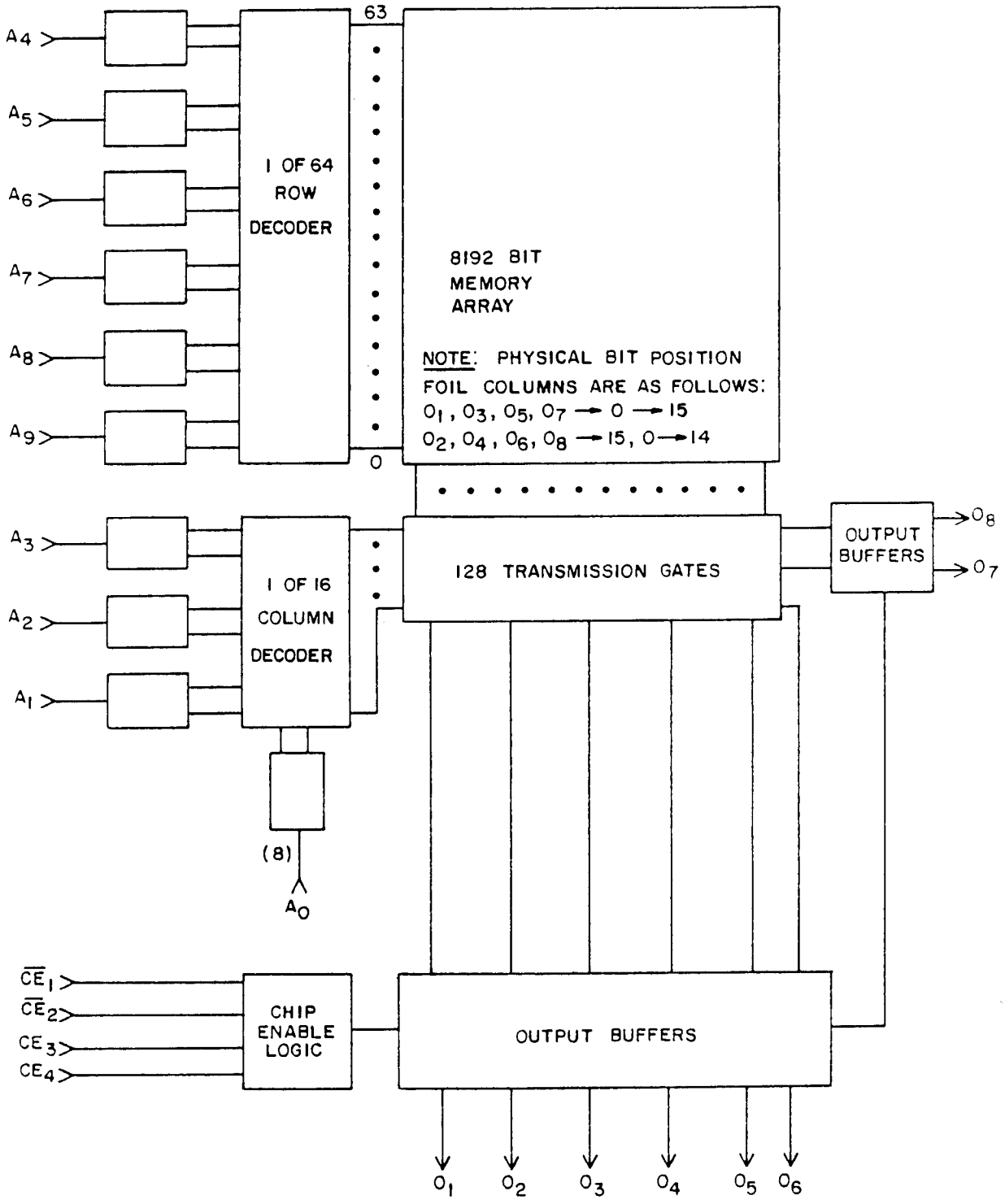
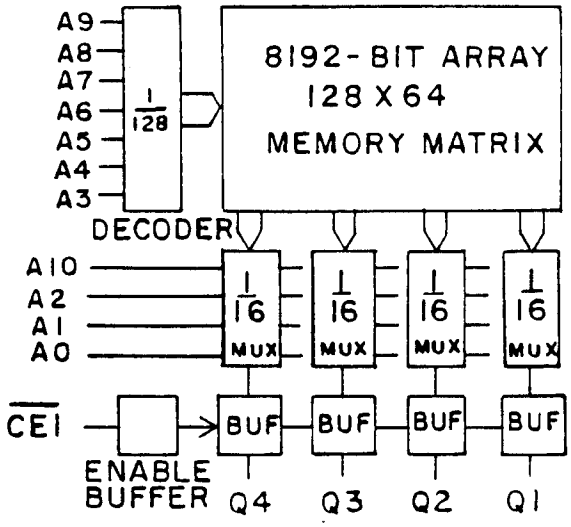


FIGURE 4. Functional block diagrams - Continued.

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Device types 01 and 02

Circuit G



Device types 03 and 04

Circuit G

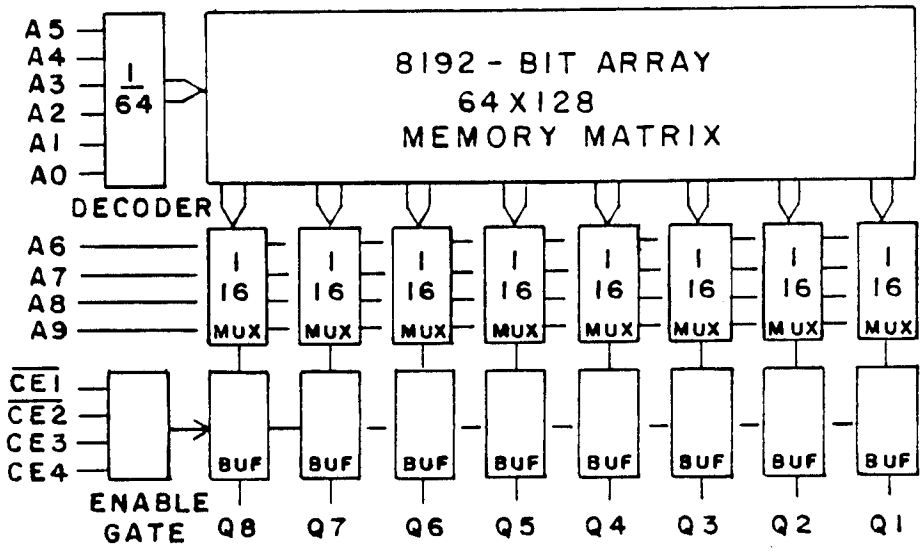


FIGURE 4. Functional block diagrams - Continued.

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Device type 04

Circuit F

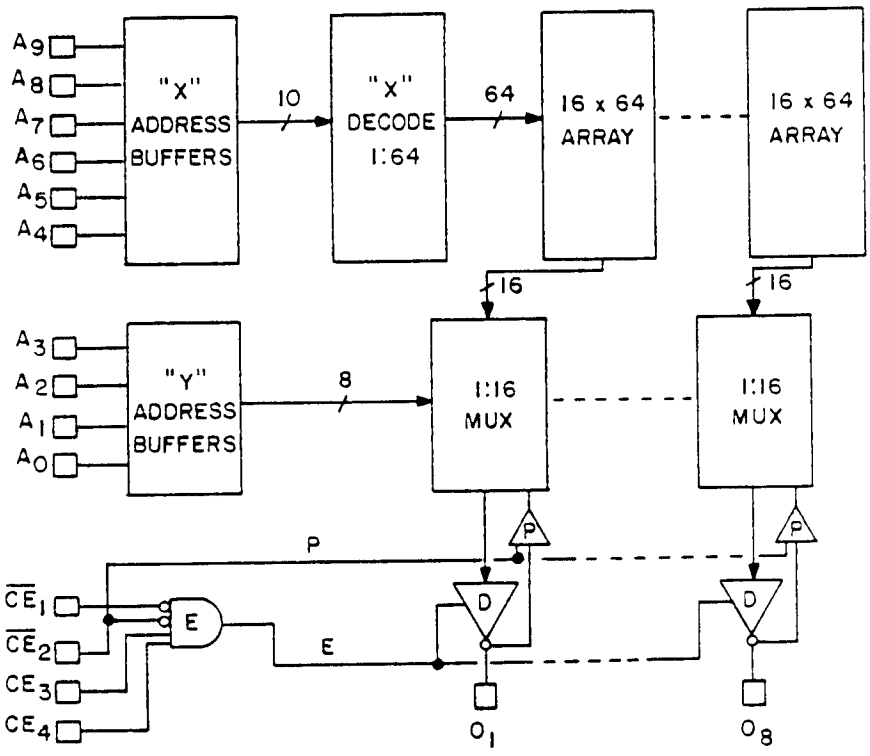


FIGURE 4. Functional block diagrams - Continued.

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Device types 03, 04, 05, and 09

Circuit C

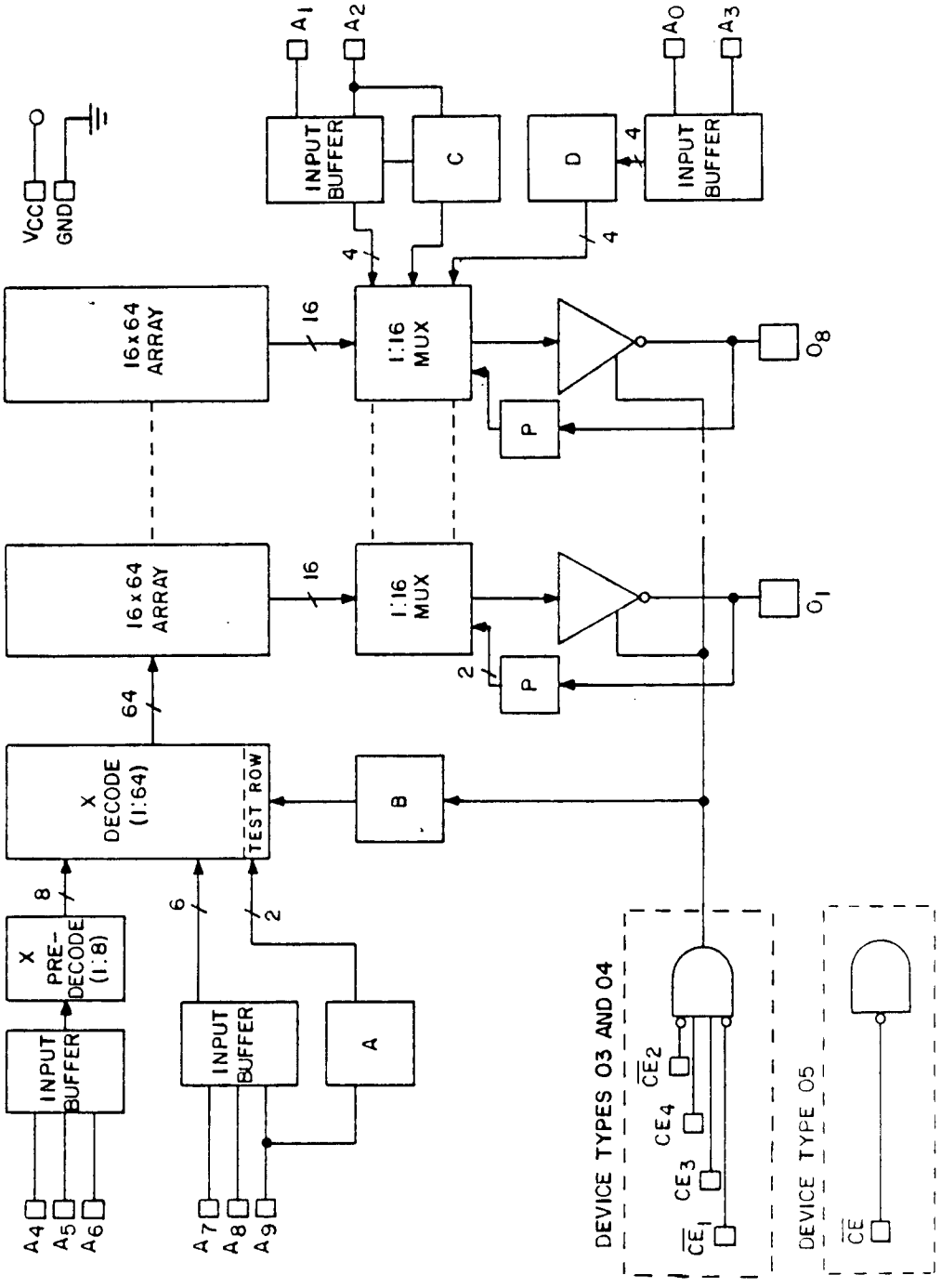


FIGURE 4. Functional block diagrams - Continued.

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Device types 03, 04, 05, 06 and 09

Circuits B and D

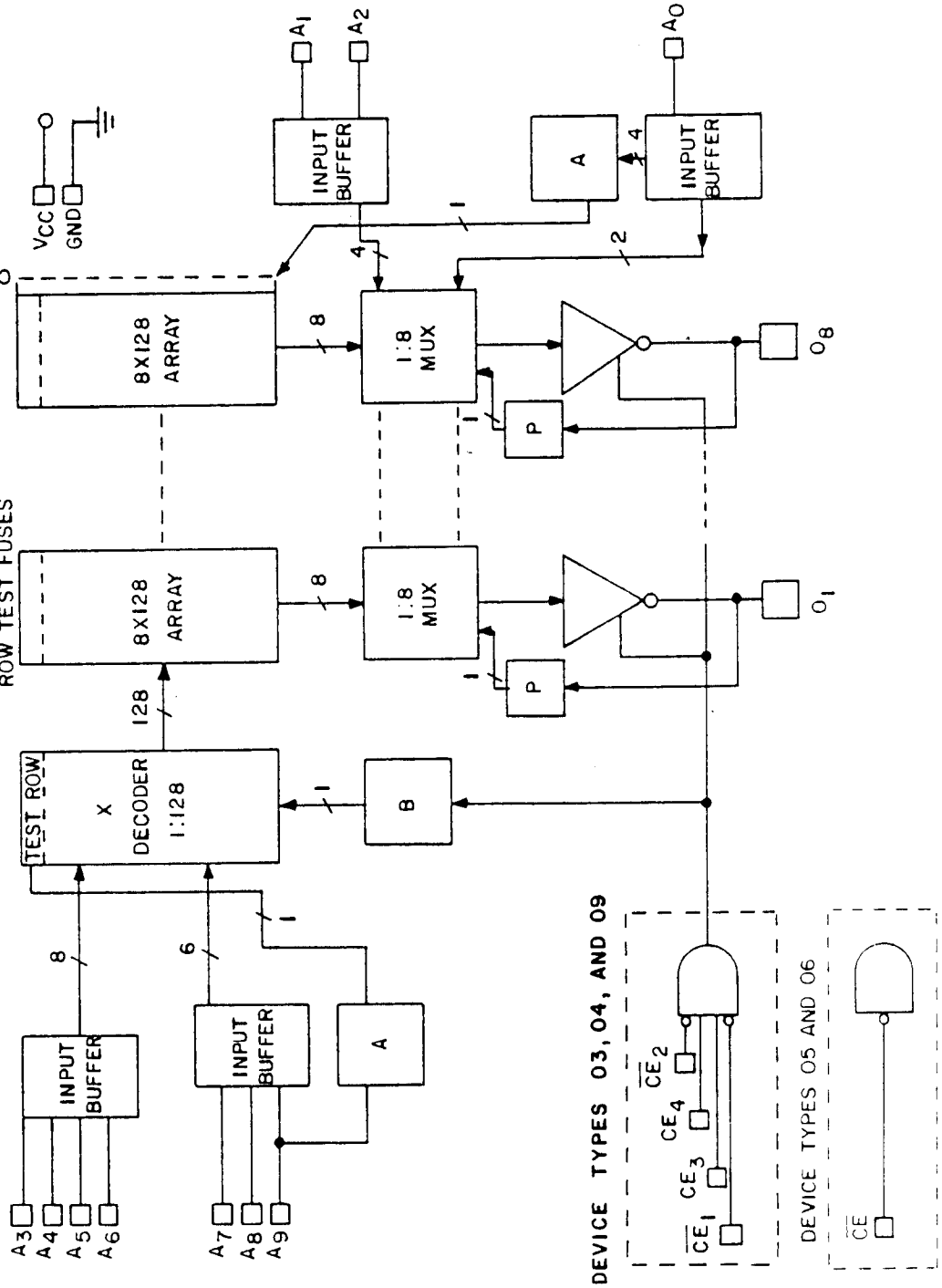
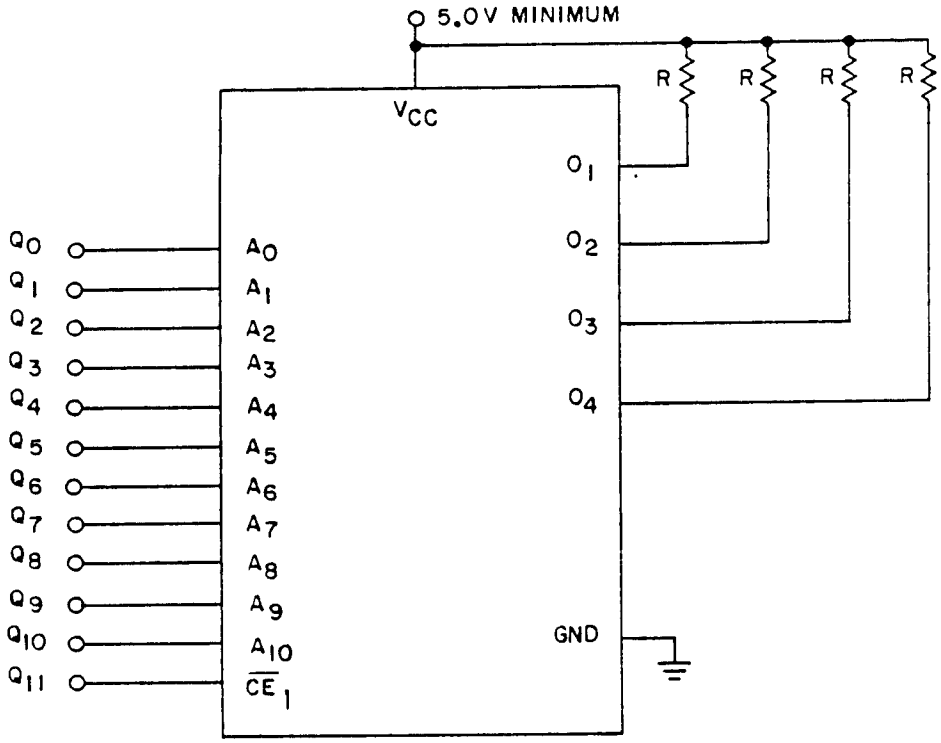


FIGURE 4. Functional block diagrams - Continued.

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Device types 01, 02, 08 and 10



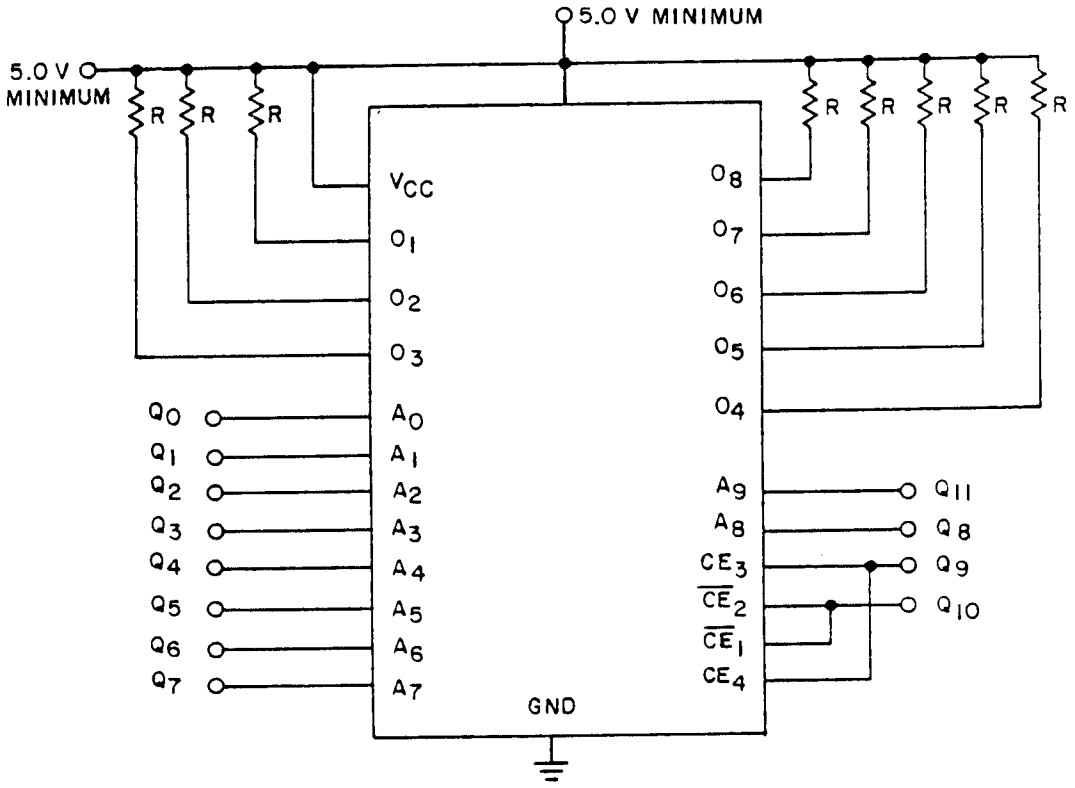
NOTES:
1. R = 360Ω ±5%. All outputs shall have separate identical loads.
2. All pulse generators have the following characteristics: V_{IL} = -1.5 V min to 0.8 V max; V_{IH} = 2.0 V min to 5.5 V max; 50% ±15% duty cycle and frequencies as specified in note 4.
3. V_{CC} shall be high enough to insure 5.0 V at the device V_{CC} terminal.
4. Input frequencies are as follows:

Input	Frequency ±50%
Q_0	F_0 = 100 kHz min
Q_1	F_1 = 1/2 F_0
Q_2	F_2 = 1/2 F_1
Q_3	F_3 = 1/2 F_2
Q_4	F_4 = 1/2 F_3
Q_5	F_5 = 1/2 F_4
Q_6	F_6 = 1/2 F_5
Q_7	F_7 = 1/2 F_6
Q_8	F_8 = 1/2 F_7
Q_9	F_9 = 1/2 F_8
Q_{10}	F_{10} = 1/2 F_9
Q_{11}	F_{11} = 1/2 F_{10}

FIGURE 5. Burn-in and steady-state life test circuit.

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Device types 03, 04 and 09



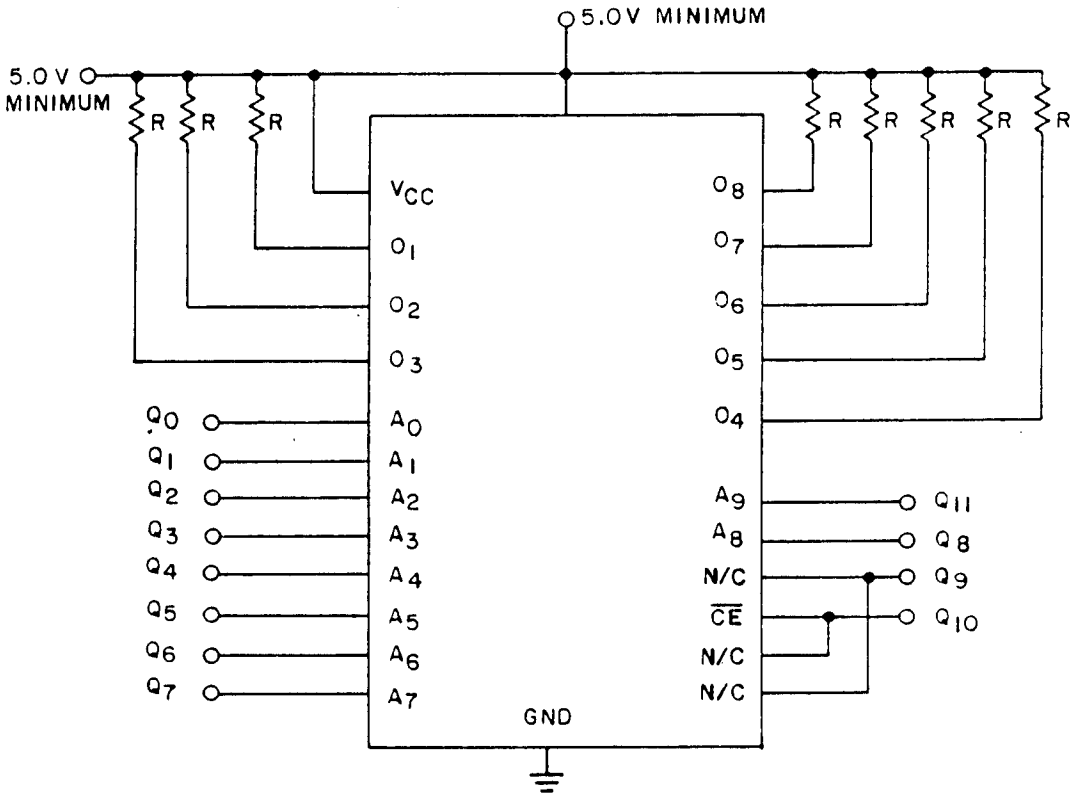
NOTES:
1. R = 560Ω ±5% (R = 300Ω ±5% for circuit F, R = 270Ω ±5% for circuit D devices). All outputs shall have separate identical loads.
2. All pulse generators have the following characteristics: V_{IL} = -1.5 V minimum to 0.8 V maximum; V_{IH} = 2.0 V minimum to 5.5 V maximum; 50% ±15% duty cycle and frequencies as specified in note 4.
3. V_{CC} shall be high enough to insure 5.0 V at the device V_{CC} terminal.
4. Input frequencies are as follows:

Input	Frequency ±50%
O_0	F_0 = 100 kHz min
O_1	F_1 = 1/2 F_0
O_2	F_2 = 1/2 F_1
O_3	F_3 = 1/2 F_2
O_4	F_4 = 1/2 F_3
O_5	F_5 = 1/2 F_4
O_6	F_6 = 1/2 F_5
O_7	F_7 = 1/2 F_6
O_8	F_8 = 1/2 F_7
O_9	F_9 = 1/2 F_8
O_{10}	F_{10} = 1/2 F_9
O_{11}	F_{11} = 1/2 F_{10}

FIGURE 5. Burn-in and steady-state life test circuit - Continued.

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Device types 05 and 06

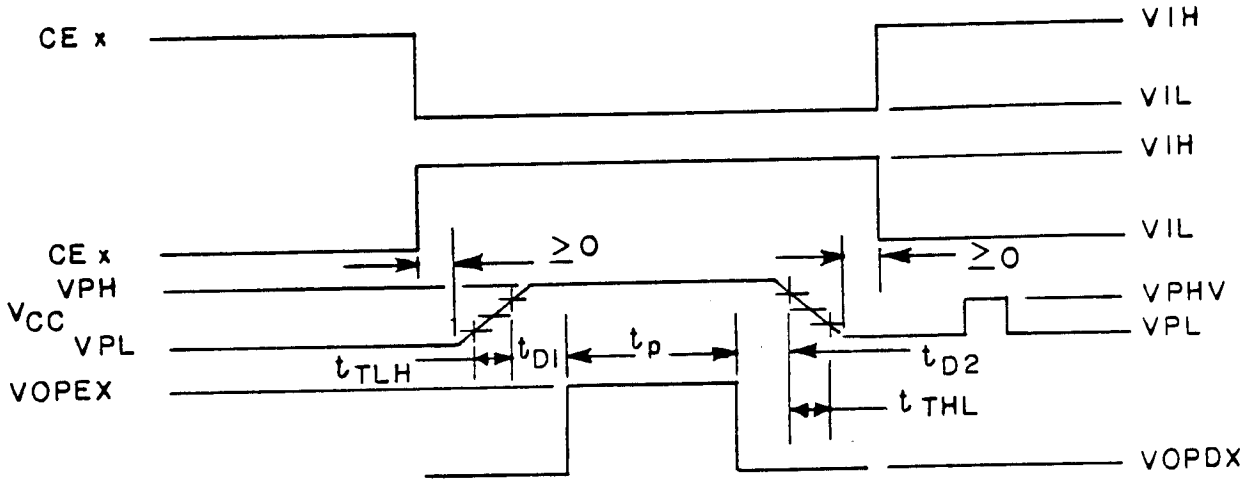


NOTES:
1. R = 560Ω ±5%. All outputs shall have separate identical loads.
2. All pulse generators have the following characteristics: V_{IL} = -1.5 V minimum to 0.8 V maximum; V_{IH} = 2.0 V minimum to 5.5 V maximum; 50% ±15% duty cycle and frequencies as specified in note 4.
3. V_{CC} shall be high enough to insure 5.0 V at the device V_{CC} terminal.
4. Input frequencies are as follows:

Input	Frequency ±50%
Q_0	F_0 = 100 kHz min
Q_1	F_1 = 1/2 F_0
Q_2	F_2 = 1/2 F_1
Q_3	F_3 = 1/2 F_2
Q_4	F_4 = 1/2 F_3
Q_5	F_5 = 1/2 F_4
Q_6	F_6 = 1/2 F_5
Q_7	F_7 = 1/2 F_6
Q_8	F_8 = 1/2 F_7
Q_9	F_9 = 1/2 F_8
Q_{10}	F_{10} = 1/2 F_9
Q_{11}	F_{11} = 1/2 F_{10}

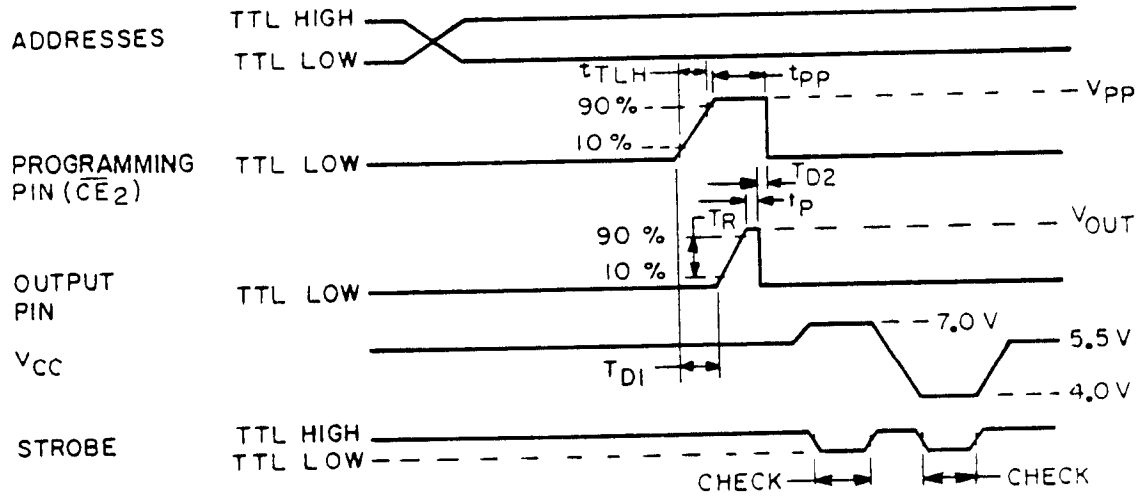
FIGURE 5. Burn-in and steady-state life test circuit - Continued.

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NOTE: All other waveform characteristics shall be as specified in table IVA.

FIGURE 6A. Programming voltage waveforms during programming for circuit A.



NOTES:
1. Output load is 0.2 mA and 12 mA during 7.0 V and 4.0 V check, respectively.
2. All other waveform characteristics shall be as specified in table IVB.

FIGURE 6B. Programming voltage waveforms during programming for circuit B (Device types 03 and 04).

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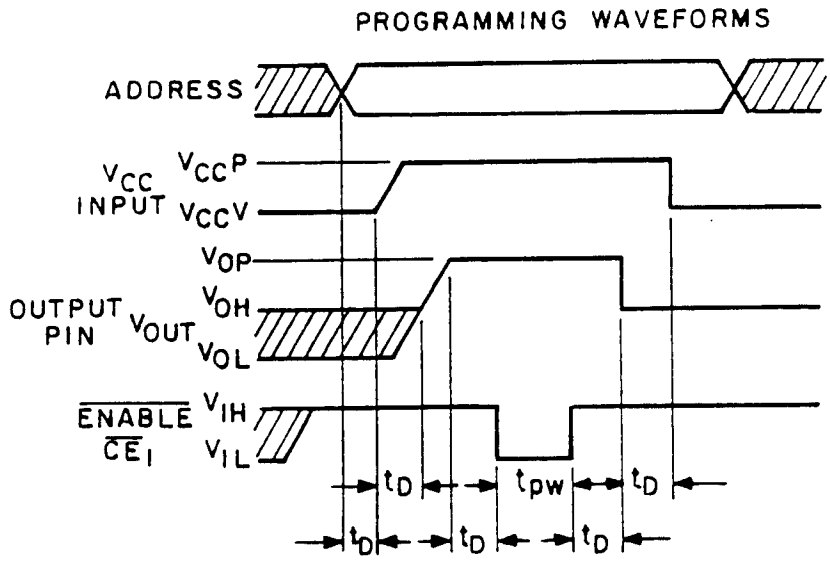
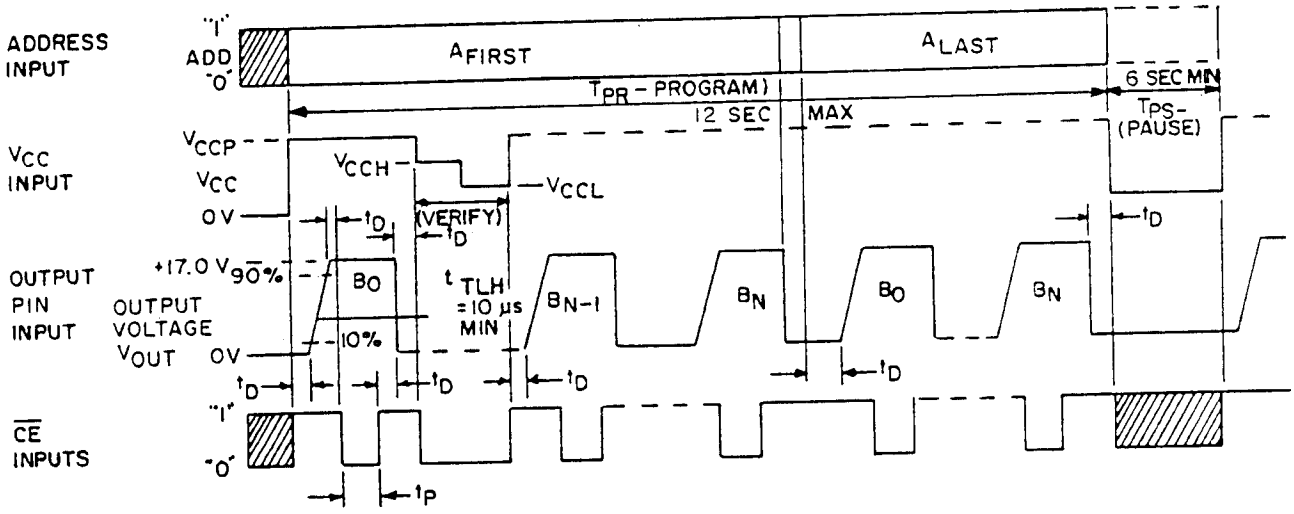


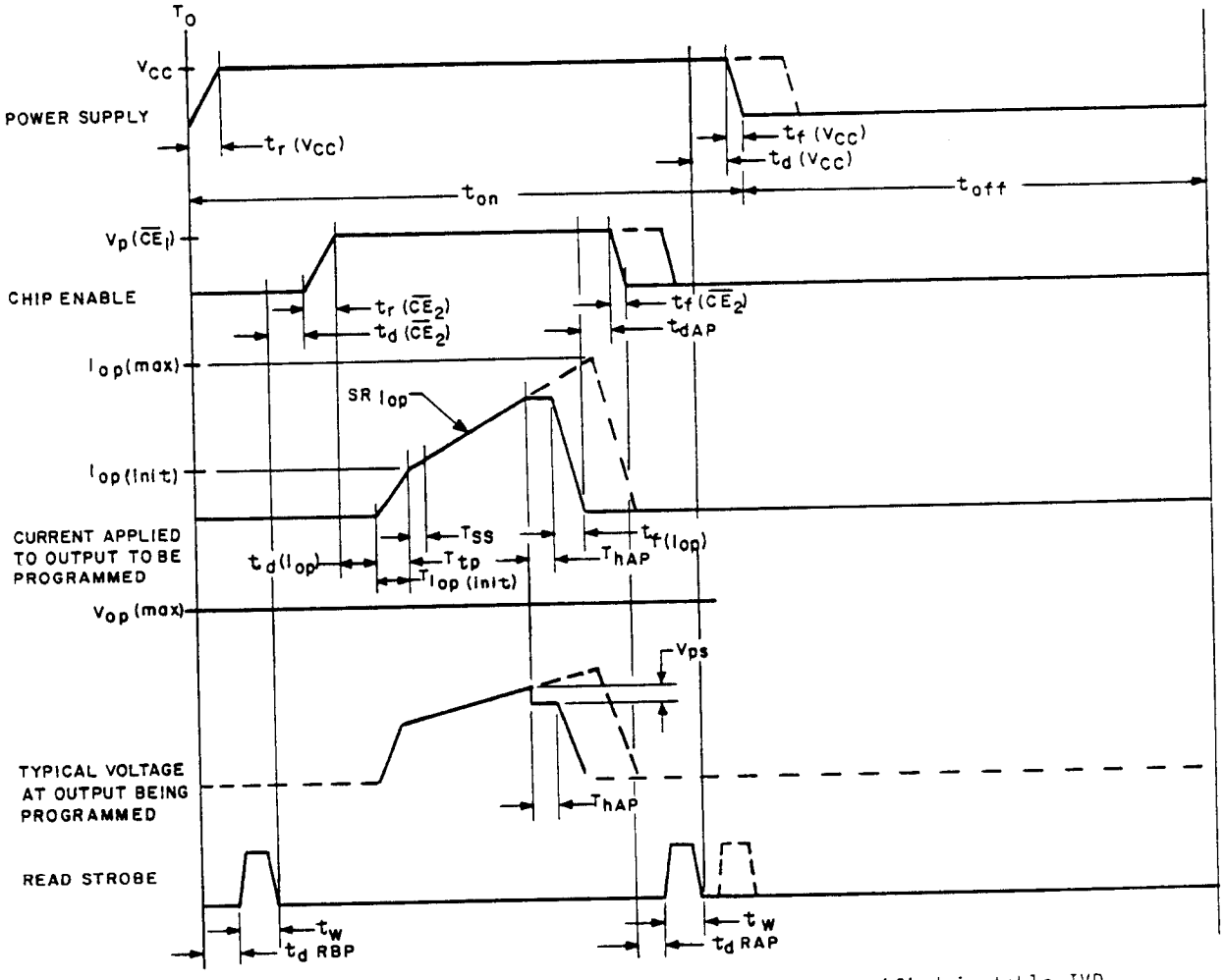
FIGURE 6B. Programming voltage waveforms during programming for circuit B (Device types 01, 02 and 08) - Continued.

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NOTE: All other waveform characteristics shall be as specified in table IVC.

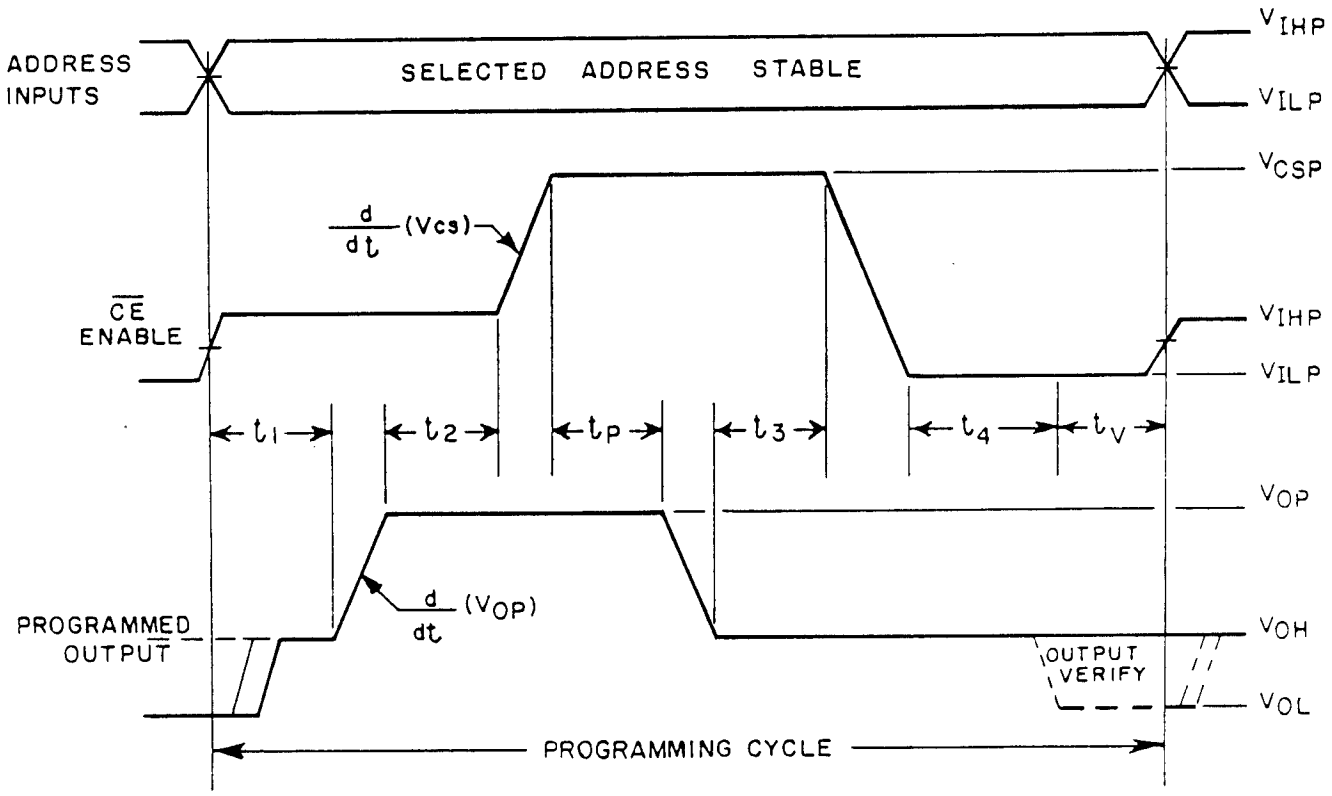
FIGURE 6C. Programming voltage waveforms during programming for circuit C.



NOTE: All other waveform characteristics shall be as specified in table IVD.

FIGURE 6D. Programming voltage waveforms during programming for circuit D.

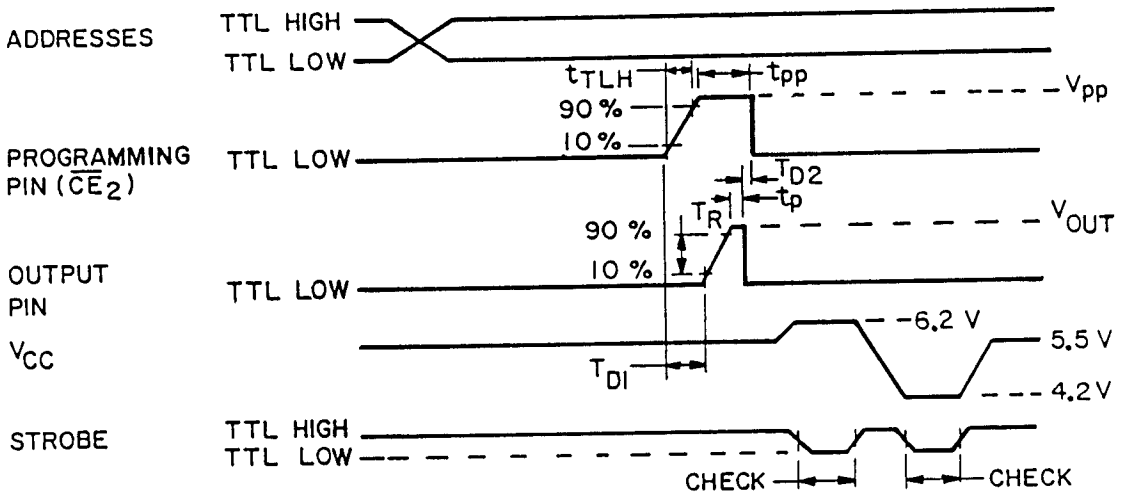
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NOTES:
1. All delays between edges are specified from completion of the first edge, not midpoints.
2. Delays t_1, t_2, t_3 and t_4 must be greater than 100 ns; maximum delays of 1 µs are recommended to minimize heating during programming.
3. During t_V the output being programmed is switched to the load R and verified.
4. Outputs not being programmed are connected to V_{ONP} through a resistor which provides output current limiting.
5. All other waveform characteristics shall be as specified in table IVE.

FIGURE 6E. Programming voltage waveforms during programming for circuit E.

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NOTES:
1. Output load is 0.2 mA and 12 mA during 6.2 V and 4.2 V check, respectively.
2. All other waveform characteristics shall be as specified in table IVF.

FIGURE 6F. Programming voltage waveforms during programming for circuit F.

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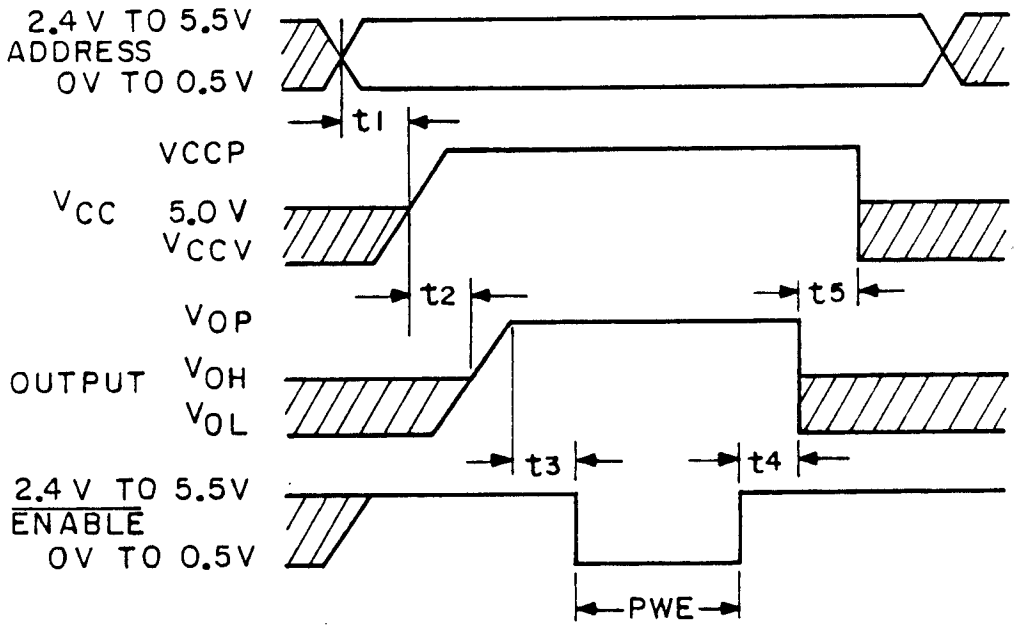
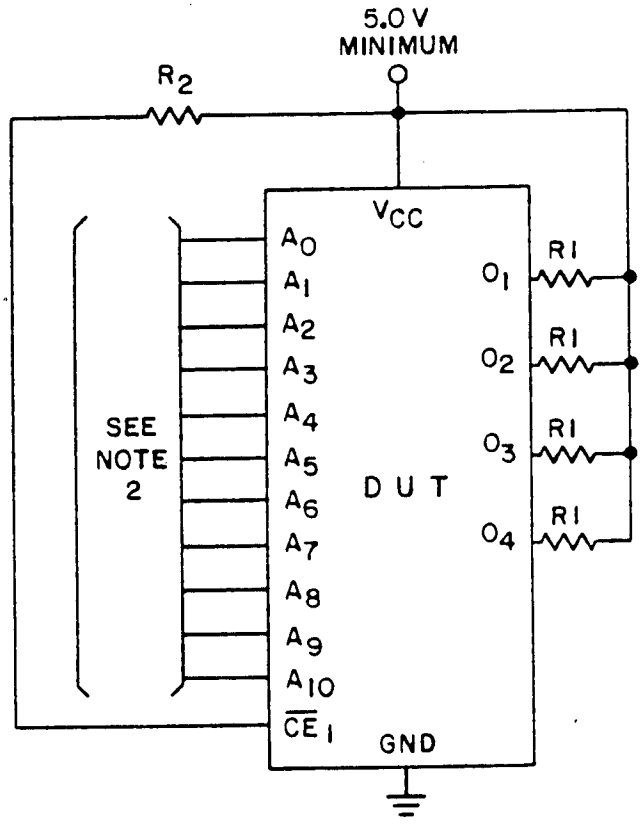


FIGURE 6G. Programming voltage waveforms during programming for circuit G.

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Device types 01, 02, 08 and 10

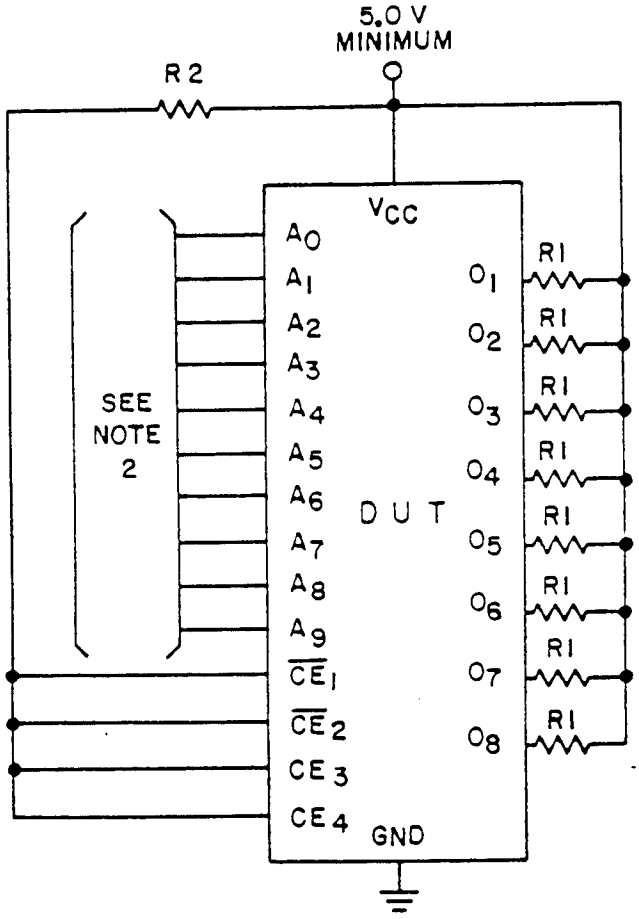


NOTES:
1. R_1 = 4.7 kΩ ±5%. All bit outputs shall have separate identical loads.
2. All address inputs shall be either high, low, or open.
3. R_2 = 1 kΩ ±5%.
4. Burn-in circuit may be used to perform this test (see 4.2d).

FIGURE 7. Freeze-out test bias configurations.

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Device types 03, 04 and 09

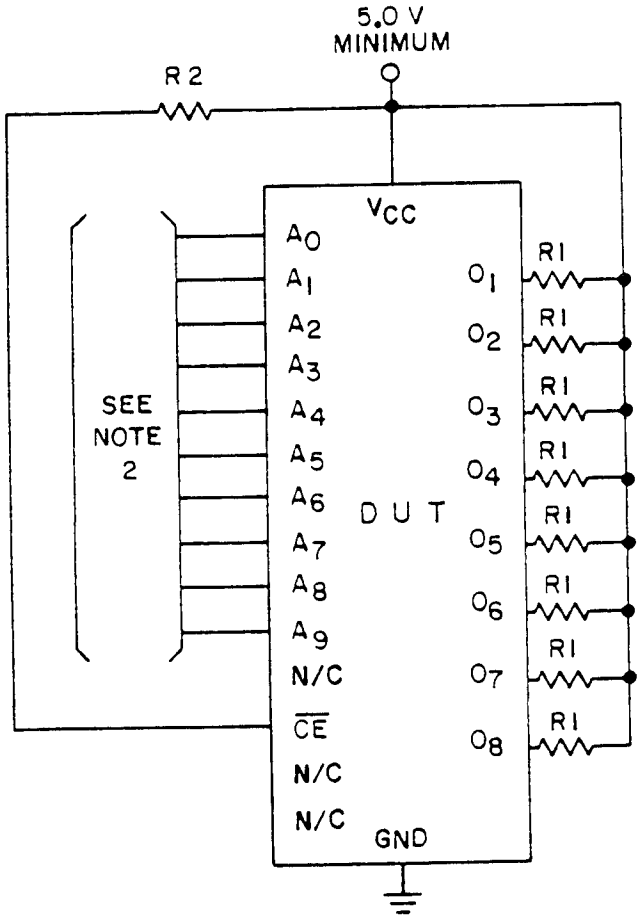


NOTES:
1. R_1 = 4.7 kΩ ±5%. All bit outputs shall have separate identical loads.
2. All address inputs shall be either high, low, or open.
3. R_2 = 1 kΩ ±5%.
4. Burn-in circuit may be used to perform this test (see 4.2d).

FIGURE 7. Freeze-out test bias configurations - Continued.

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Device types 05 and 06

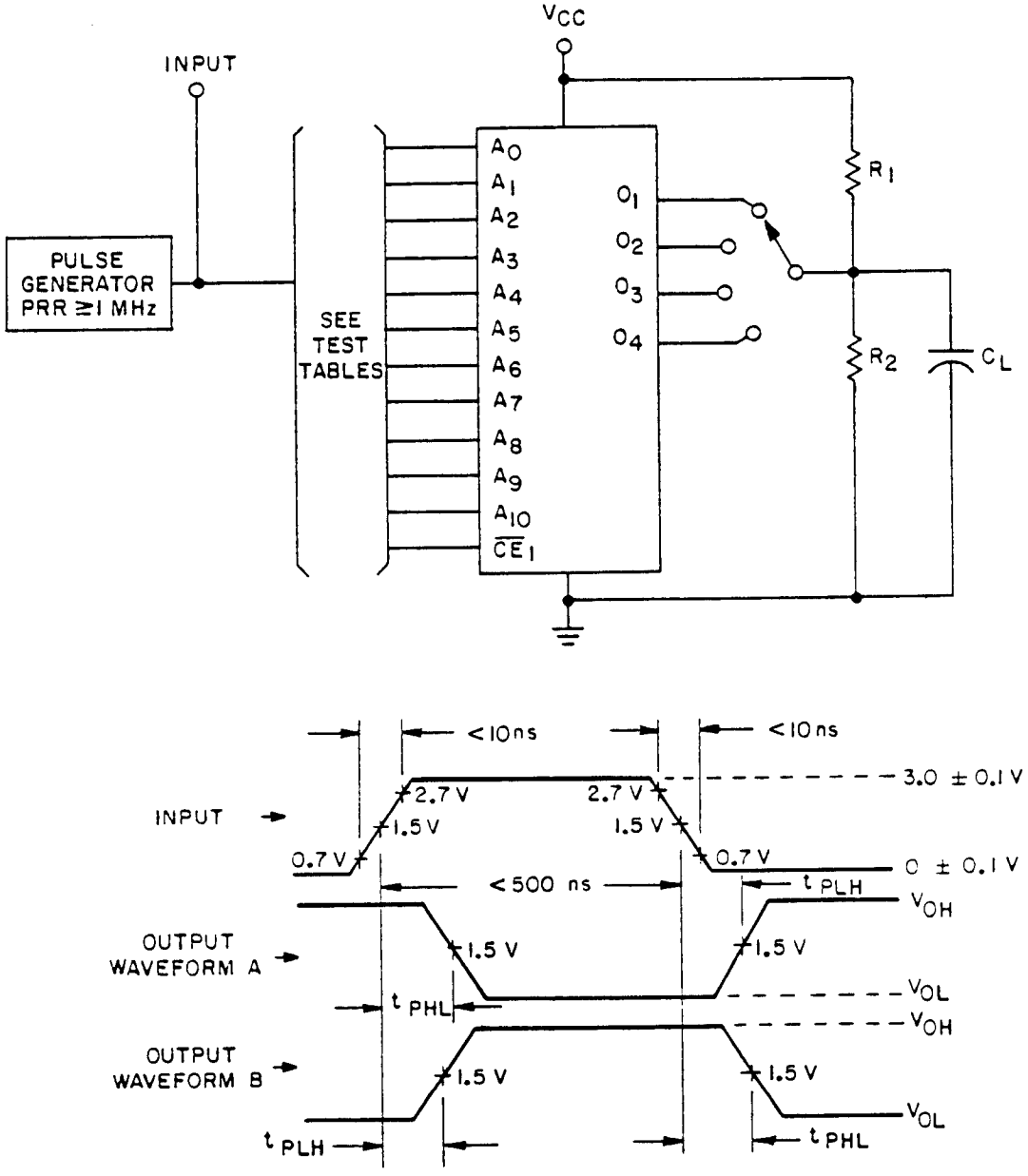


NOTES:
1. R_1 = 4.7 kΩ ±5%. All bit outputs shall have separate identical loads.
2. All address inputs shall be either high, low, or open.
3. R_2 = 1 kΩ ±5%.
4. Burn-in circuit may be used to perform this test (see 4.2d).

FIGURE 7. Freeze-out test bias configurations - Continued.

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Device types 01, 02, 08 and 10

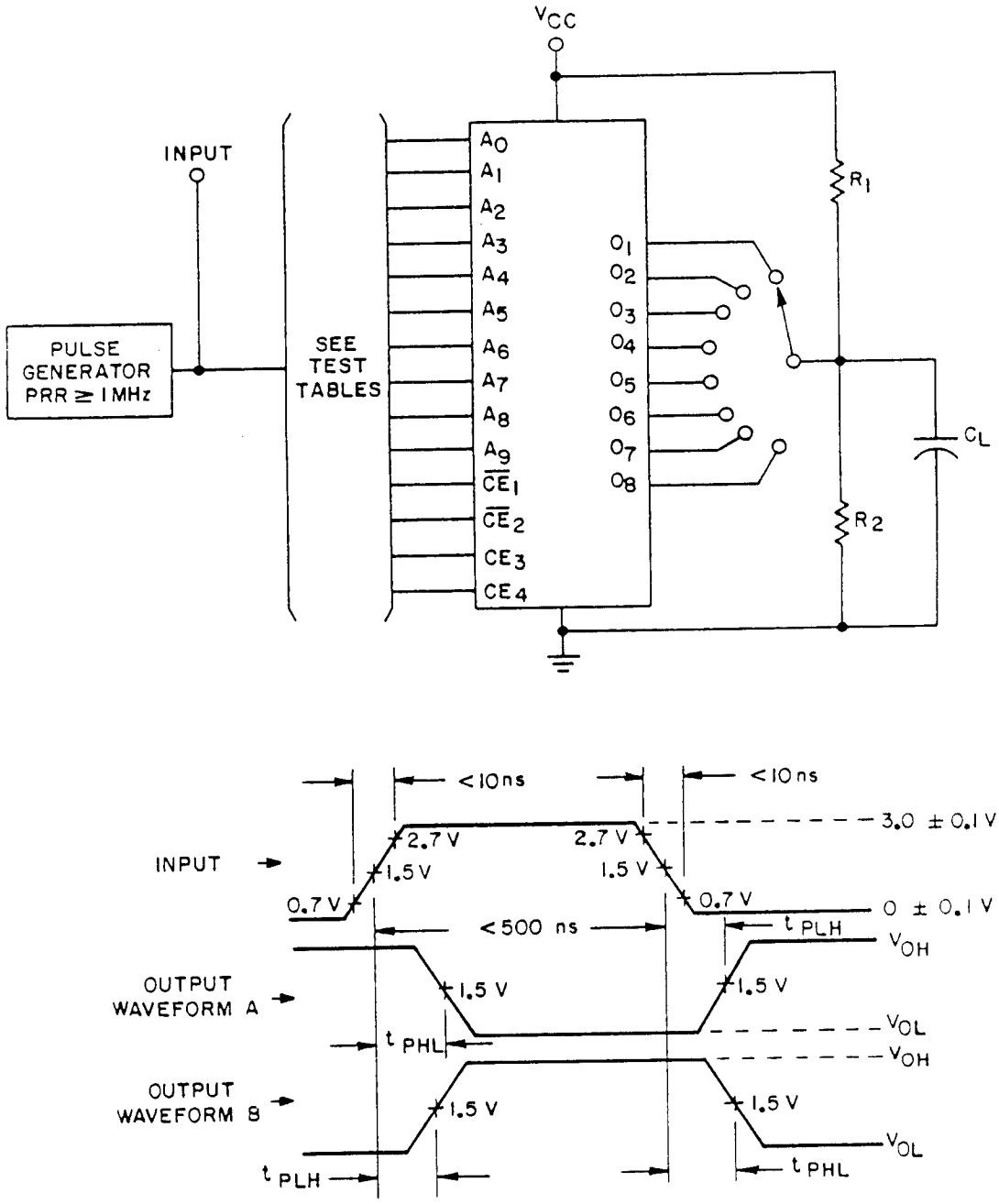


NOTES:
1. Test table for devices programmed in accordance with an altered item drawing may be replaced by the equivalent tests which apply to the specific program configuration for the resulting read-only memory.
2. C_L = 30 pF minimum, including jig and probe capacitance; R_1 = 330Ω ±25% and R_2 = 680Ω ±20%.
3. Outputs may be under load simultaneously.

FIGURE 8. Switching time test circuits.

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Device types 03, 04 and 09

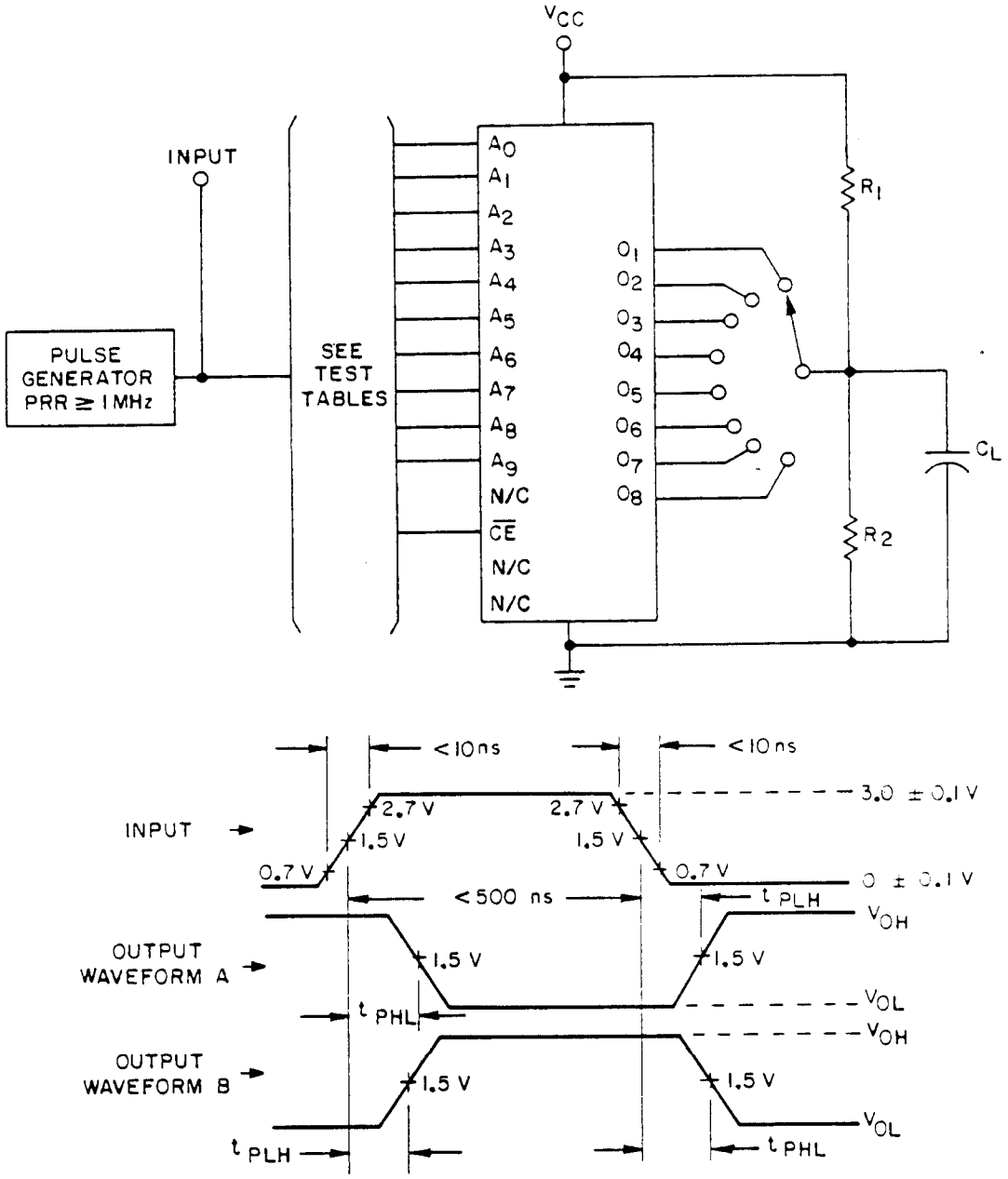


NOTES:
1. Test table for devices programmed in accordance with an altered item drawing may be replaced by the equivalent tests which apply to the specific program configuration for the resulting read-only memory.
2. C_L = 30 pF minimum, including jig and probe capacitance; R_1 = 330Ω ±25% and R_2 = 680Ω ±20%.
3. Outputs may be under load simultaneously.

FIGURE 8. Switching time test circuits - Continued.

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Device types 05 and 06



NOTES:
1. Test table for devices programmed in accordance with an altered item drawing may be replaced by the equivalent tests which apply to the specific program configuration for the resulting read-only memory.
2. C_L = 30 pF minimum, including jig and probe capacitance; R_1 = 330Ω ±25% and R_2 = 680Ω ±20%.
3. Outputs may be under load simultaneously.

FIGURE 8. Switching time test circuits - Continued.

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TABLE III. Group A inspection for device type 01.

Terminal conditions (outputs not designated are open or resistive, coupled to GND or voltage, inputs not designated are high ≥2.0 V, low ≤0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases V,X	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	Meas. terminal	Test limits Min	Test limits Max	Unit
			Test no.	A_6	A_5	A_4	A_3	A_0	A_1	A_2	A_{10}	GND	$\overline{CE}_1$	O_4	O_3	O_2	O_1	A_9	A_8	A_7	V_{CC}				
1 T_C = +25°C	V_{IC}		1	-10 mA	-10 mA	-10 mA	-10 mA	-10 mA	-10 mA	-10 mA	-10 mA	GND	-10 mA					-10 mA	-10 mA	-10 mA	4.5 V	A_6 A_5 A_4 A_3 A_0 A_1 A_2 A_{10} $\overline{CE}_1$ A_9 A_8 A_7		-1.5	V
	V_{OL}	3007	13 14 15 16	1/ 2/ " "	1/ " "	1/ " "	1/ " "	1/ " "	1/ " "	1/ " "	1/ " "	" " " "	0.5 V " " "	3/	3/	3/	3/	1/ " "	1/ " "	1/ " "	" " " "	O_4 O_3 O_2 O_1		0.5 " " "	" " " "
	I_{IL}	3009	17 18 19 20 21 22 23 24 25 26 27 28	0.5 V	0.5 V	0.5 V	0.5 V	0.5 V	0.5 V	0.5 V	0.5 V	" " " " " " " " " " " "	0.5 V					0.5 V	0.5 V	0.5 V	5.5 V	A_6 A_5 A_4 A_3 A_0 A_1 A_2 A_{10} $\overline{CE}_1$ A_9 A_8 A_7	-1.0	-250	µA
	I_{IH}	3010	29 30 31 32 33 34 35 36 37 38 39 40	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	" " " " " " " " " " " "	5.5 V					5.5 V	5.5 V	5.5 V	" " " " " " " " " " " "	A_6 A_5 A_4 A_3 A_0 A_1 A_2 A_{10} A_9 A_8 A_7 $\overline{CE}_1$		50	"
	I_{CEX}		41 42 43 44									" " " "	" " " "	5.2 V	5.2 V	5.2 V	5.2 V				" " " "	O_4 O_3 O_2 O_1		100 " " "	" " " "
	I_{CC}	3005	45	GND	GND	GND	GND	GND	GND	GND	GND	"	GND					GND	GND	GND	"	V_{CC}		170	mA
2	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = +125°C and V_{IC} tests are omitted.																								
3	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = -55°C and V_{IC} tests are omitted.																								
7 T_C = +25°C	Functional tests	4/	46	4/	4/	4/	4/	4/	4/	4/	4/	GND	4/	4/	4/	4/	4/	4/	4/	4/	4/	Outputs		4/	

See footnotes at end of table.

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TABLE III. Group A inspection for device type 01 - Continued.

Terminal conditions (outputs not designated are open or resistive, coupled to GND or voltage, inputs not designated are high ≥2.0 V, low ≤0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases V,X Test no.	1 A_6	2 A_5	3 A_4	4 A_3	5 A_0	6 A_1	7 A_2	8 A_{10}	9 GND	10 $\overline{CE}_1$	11 O_4	12 O_3	13 O_2	14 O_1	15 A_9	16 A_8	17 A_7	18 V_{CC}	Meas. terminal	Test limits Min	Test limits Max	Unit
8	Same tests, terminal conditions, and limits as for subgroup 7, except T_C = +125°C and -55°C.																								
9 T_C = +25°C	t_{PHL1}	GALPAT Fig. 8	47	5/	5/	5/	5/	5/	5/	5/	5/	GND	GND	6/	6/	6/	6/	5/	5/	5/	5/	Outputs		125	ns
	t_{PLH1}	GALPAT Fig. 8	48	5/	5/	5/	5/	5/	5/	5/	5/	"	GND	"	"	"	"	5/	5/	5/	5/	"		125	"
	t_{PHL2}	Sequential Fig. 8	49	7/	7/	7/	7/	7/	7/	7/	7/	"	7/	"	"	"	"	7/	7/	7/	7/	"		60	"
	t_{PLH2}	Sequential Fig. 8	50	7/	7/	7/	7/	7/	7/	7/	7/	"	7/	"	"	"	"	7/	7/	7/	7/	"		60	"
																								8/	
10	Same tests, terminal conditions, and limits as for subgroup 9, except T_C = +125°C.																								
11	Same tests, terminal conditions, and limits as for subgroup 9, except T_C = -55°C.																								

See footnotes at end of table.

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TABLE III. Group A inspection for device types 02, 08, and 10.

Terminal conditions (outputs not designated are open or resistive, coupled to GND or voltage, inputs not designated are high ≥2.0 V, low ≤0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases V, X Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	Meas. terminal	Test limits Min	Test limits Max	Unit
				A_6	A_5	A_4	A_3	A_0	A_1	A_2	A_{10}	GND	$\overline{CE}_1$	O_4	O_3	O_2	O_1	A_9	A_8	A_7	V_{CC}				
1 T_C = +25°C	V_{IC}		1	-10 mA								GND									4.5 V	A_6		-1.5	V
			2		-10 mA							"									"	A_5		"	"
			3			-10 mA						"									"	A_4		"	"
			4				-10 mA					"									"	A_3		"	"
			5					-10 mA				"									"	A_0		"	"
			6						-10 mA			"									"	A_1		"	"
			7							-10 mA		"									"	A_2		"	"
			8								-10 mA	"									"	A_{10}		"	"
			9									"	-10 mA								"	$\overline{CE}_1$		"	"
			10									"						-10 mA			"	A_9		"	"
			11									"							-10 mA		"	A_8		"	"
			12									"								-10 mA	"	A_7		"	"
	V_{OL}	3007	13	1/ 2/	1/	1/	1/	1/ 9/	1/	1/	1/	"	0.5 V	3/	3/	3/	3/	1/	1/	1/	"	O_4		0.5	"
			14	"	"	"	"	"	"	"	"	"	"					"	"	"	"	O_3		"	"
			15	"	"	"	"	"	"	"	"	"	"					"	"	"	"	O_2		"	"
			16	"	"	"	"	"	"	"	"	"	"					"	"	"	"	O_1		"	"
	V_{OH}	3006	17	1/10/11/	1/9/10/	1/ 12/	"	1/13/	"	"	1/10/12/	"	"	-2 mA	-2 mA	-2 mA	-2 mA	1/ 9/	1/13/	"	"	O_4	2.4		"
			18	"	"	"	"	"	"	"	"	"	"					"	"	"	"	O_3	"		"
			19	"	"	"	"	"	"	"	"	"	"					"	"	"	"	O_2	"		"
			20	"12/	"	"	"	"	"	"	"	"	"					"	"	"	"	O_1	"		"
	I_{IL}	3009	21	0.5 V								"									5.5 V	A_6	-1.0	-250	μA
			22		0.5 V							"									"	A_5	"	"	"
			23			0.5 V						"									"	A_4	"	"	"
			24				0.5 V					"									"	A_3	"	"	"
			25					0.5 V				"									"	A_0	"	"	"
			26						0.5 V			"									"	A_1	"	"	"
			27							0.5 V		"									"	A_2	"	"	"
			28								0.5 V	"									"	A_{10}	"	"	"
			29									"	0.5 V								"	$\overline{CE}_1$	"	"	"
			30									"						0.5 V			"	A_9	"	"	"
			31									"							0.5 V		"	A_8	"	"	"
			32									"								0.5 V	"	A_7	"	"	"
	I_{IH}	3010	33	5.5 V								"									"	A_6		50	μA
			34		5.5 V							"									"	A_5		"	"
			35			5.5 V						"									"	A_4		"	"
			36				5.5 V					"									"	A_3		"	"
			37					5.5 V				"									"	A_0		"	"
			38						5.5 V			"									"	A_1		"	"
			39							5.5 V		"									"	A_2		"	"
			40								5.5 V	"									"	A_{10}		"	"
			41									"						5.5 V			"	A_9		"	"
			42									"							5.5 V		"	A_8		"	"
			43									"								5.5 V	"	A_7		"	"
			44									"	5.5 V								"	$\overline{CE}_1$		"	"
	I_{OHZ}		45									"	14/	5.2 V	5.2 V	5.2 V	5.2 V				"	O_4		100	"
			46									"	"								"	O_3		"	"
			47									"	"								"	O_2		"	"
			48									"	"								"	O_1		"	"
	I_{OLZ}		49									"	"	0.5 V	0.5 V	0.5 V	0.5 V				"	O_4		-100	"
			50									"	"								"	O_3		"	"
			51									"	"								"	O_2		"	"
			52									"	"								"	O_1		"	"
	I_{CC}	3005	53	GND	GND	GND	GND	GND	GND	GND	GND	"	GND					GND	GND	GND	"	V_{CC}		170	mA

See footnotes at end of table.

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TABLE III. Group A inspection for device types 02, 08, and 10 - Continued.

Terminal conditions (outputs not designated are open or resistive, coupled to GND or voltage, inputs not designated are high ≥2.0 V, low ≤0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases V, X Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	Meas. terminal	Test limits Min	Test limits Max	Unit
				A_6	A_5	A_4	A_3	A_0	A_1	A_2	A_{10}	GND	$\overline{CE}_1$	O_4	O_3	O_2	O_1	A_9	A_8	A_7	V_{CC}				
1 T_C = +25°C	I_{OS}	3011	54	1/10/11/	1/9/10/	1/	1/	1/13/	1/	1/	1/ 10/	GND	0.5 V	GND	GND			1/ 9/	1/13/	1/	5.5 V	O_4	-15	-100	mA
			55	"	"	"	"	"	"	"	"	"	"			GND		"	"	"	"	O_3	"	"	"
			56	"	"	"	"	"	"	"	"	"	"				GND	"	"	"	"	O_2	"	"	"
			57	"12/	"	"12/	"	"	"	"	"12/	"	"					"	"	"	"	O_1	"	"	"
2	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = +125°C and V_{IC} tests are omitted.																								
3	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = -55°C and V_{IC} tests are omitted.																								
7 T_C = +25°C	Functional tests	4/	58	4/	4/	4/	4/	4/	4/	4/	4/	GND	4/	4/	4/	4/	4/	4/	4/	4/	4/	Outputs		4/	
8	Same tests, terminal conditions, and limits as for subgroup 7, except T_C = +125°C and T_C = -55°C.																								
9 T_C = +25°C	t_{PHL1}	GALPAT Fig. 8	59	5/	5/	5/	5/	5/	5/	5/	5/	GND	GND	6/	6/	6/	6/	5/	5/	5/	5/	Outputs		8/	ns
	t_{PLH1}	GALPAT Fig. 8	60	5/	5/	5/	5/	5/	5/	5/	5/	"	GND	"	"	"	"	5/	5/	5/	5/	"		"	"
	t_{PHL2}	Sequential Fig. 8	61	7/	7/	7/	7/	7/	7/	7/	7/	"	7/	"	"	"	"	7/	7/	7/	7/	"		"	"
	t_{PLH2}	Sequential Fig. 8	62	7/	7/	7/	7/	7/	7/	7/	7/	"	7/	"	"	"	"	7/	7/	7/	7/	"		"	"
10	Same tests, terminal conditions, and limits as for subgroup 9, except T_C = +125°C.																								
11	Same tests, terminal conditions, and limits as for subgroup 9, except T_C = -55°C.																								

See footnotes at end of table.

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TABLE III. Group A inspection for device type 03.

Terminal conditions (outputs not designated are open or resistive coupled to GND or voltage; inputs not designated are high ≥2.0 V, low ≤0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases J,K	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	Measured terminal	Test limits		Unit
			Test no.	A_7	A_6	A_5	A_4	A_3	A_2	A_1	A_0	O_1	O_2	O_3	GND	O_4	O_5	O_6	O_7	O_8	$\overline{CE}_4$	$\overline{CE}_3$	$\overline{CE}_2$	$\overline{CE}_1$	A_9	A_8	V_{CC}		Min	Max	
1 T_C = +25°C	V_{IC}		1 2 3 4 5 6 7 8 9 10 11 12 13 14	-10 mA	-10 mA	-10 mA	-10 mA	-10 mA	-10 mA	-10 mA	-10 mA				"						-10 mA	-10 mA	-10 mA	-10 mA	-10 mA	-10 mA	4.5 V " " " " " " " " " " " " "	A_7 A_6 A_5 A_4 A_3 A_2 A_1 A_0 $\overline{CE}_4$ $\overline{CE}_3$ $\overline{CE}_2$ $\overline{CE}_1$ A_9 A_8		-1.5 " " " " " " " " " " " " "	V " " " " " " " " " " " " "
	V_{OL}	3007	15 16 17 18 19 20 21 22	1/ 2/ 15/7 " " " " " " "	1/ " " " " " " "	1/ " " " " " " "	1/ " " " " " " "	1/ " " " " " " "	1/ " " " " " " "	1/ " " " " " " "	1/ " " " " " " "	8 mA	8 mA	8 mA	" " " " " " " "	8 mA	8 mA	8 mA	8 mA	8 mA	5.5 V " " " " " " "	5.5 V " " " " " " "	0.5 V " " " " " " "	0.5 V " " " " " " "	1/ 15/ " " " " " " "	1/ " " " " " " "	" " " " " " " "	O_1 O_2 O_3 O_4 O_5 O_6 O_7 O_8		0.5 " " " " " " "	" " " " " " " "
	I_{IL}	3009	23 24 25 26 27 28 29 30 31 32 33 34 35 36	0.5 V	0.5 V	0.5 V	0.5 V	0.5 V	0.5 V	0.5 V	0.5 V				"						0.5 V	0.5 V	0.5 V	0.5 V	0.5 V	0.5 V	5.5 V " " " " " " " " " " " " "	A_7 A_6 A_5 A_4 A_3 A_2 A_1 A_0 $\overline{CE}_4$ $\overline{CE}_3$ $\overline{CE}_2$ $\overline{CE}_1$ A_9 A_8	-1.0 " " " " " " " " " " " " "	-250 " " " " " " " " " " " " "	µA " " " " " " " " " " " " "
	I_{IH1}	3010	37 38 39 40 41 42 43 44 45 46 47 48 49	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V				"						5.5 V	5.5 V		5.5 V	5.5 V	5.5 V	" " " " " " " " " " " " "	A_7 A_6 A_5 A_4 A_3 A_2 A_1 A_0 $\overline{CE}_4$ $\overline{CE}_3$ $\overline{CE}_1$ A_9 A_8		50 " " " " " " " " " " " "	" " " " " " " " " " " " "
	I_{IH2} 4/ 16/ 17/		50												"								4.5 V				"	$\overline{CE}_2$		100	"
	I_{CEX}		51 52 53 54									5.2 V	5.2 V	5.2 V	" " " "	5.2 V					0.5 V " " "	0.5 V " " "	5.5 V " " "	5.5 V " " "			" " " "	O_1 O_2 O_3 O_4		" " " "	" " " "

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TABLE III. Group A inspection for device type 03 - Continued.

Terminal conditions (outputs not designated are open or resistive coupled to GND or voltage; inputs not designated are high ≥2.0 V, low ≤0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases J,K	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	Measured terminal	Test limits		Unit
			Test no.	A_7	A_6	A_5	A_4	A_3	A_2	A_1	A_0	O_1	O_2	O_3	GND	O_4	O_5	O_6	O_7	O_8	$\overline{CE}_4$	$\overline{CE}_3$	$\overline{CE}_2$	$\overline{CE}_1$	A_9	A_8	V_{CC}		Min	Max	
1 T_C = +25°C	I_{CEX}		55 56 57 58												GND " " "		5.2 V	5.2 V	5.2 V	5.2 V	0.5 V " "	0.5 V " "	5.5 V " "	5.5 V " "			5.5 V " "	O_5 O_6 O_7 O_8		100 " "	μA " "
	I_{CC}	3005	59	GND	GND	GND	GND	GND	GND	GND	GND	GND			"						GND	GND	GND	GND	GND	GND	"	V_{CC}		185	mA
2	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = +125°C and V_{IC} tests are omitted.																														
3	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = -55°C and V_{IC} tests are omitted.																														
7 T_C = +25°C	Functional tests	4/	60	4/	4/	4/	4/	4/	4/	4/	4/	4/	4/	4/	GND	4/	4/	4/	4/	4/	4/	4/	4/	4/	4/	4/	4/	Outputs		4/	
8	Same tests, terminal conditions, and limits as for subgroup 7, except T_C = +125°C and T_C = -55°C.																														
9 T_C = +25°C	t_{PHL1}	GALPAT Fig. 8	61	5/	5/	5/	5/	5/	5/	5/	5/	6/	6/	6/	GND	6/	6/	6/	6/	6/	5.5 V	5.5 V	GND	GND	5/	5/	5/	Outputs		8/	ns
	t_{PLH1}	GALPAT Fig. 8	62	5/	5/	5/	5/	5/	5/	5/	5/	"	"	"	"	"	"	"	"	"	5.5 V	5.5 V	GND	GND	5/	5/	5/	"		"	"
	t_{PHL2}	Sequential Fig. 8	63	7/	7/	7/	7/	7/	7/	7/	7/	"	"	"	"	"	"	"	"	"	7/	7/	7/	7/	7/	7/	7/	"		"	"
	t_{PLH2}	Sequential Fig. 8	64	7/	7/	7/	7/	7/	7/	7/	7/	"	"	"	"	"	"	"	"	"	7/	7/	7/	7/	7/	7/	7/	"		"	"
10	Same tests, terminal conditions, and limits as for subgroup 9, except T_C = +125°C.																														
11	Same tests, terminal conditions, and limits as for subgroup 9, except T_C = -55°C.																														

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TABLE III. Group A inspection for device types 04 and 09.

Terminal conditions (outputs not designated are open or resistive coupled to GND or voltage; inputs not designated are high ≥2.0 V, low ≤0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases J,K	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	Measured terminal	Test limits Min	Test limits Max	Unit
			Test no.	A_7	A_6	A_5	A_4	A_3	A_2	A_1	A_0	O_1	O_2	O_3	GND	O_4	O_5	O_6	O_7	O_8	$\overline{CE}_4$	$\overline{CE}_3$	$\overline{CE}_2$	$\overline{CE}_1$	A_9	A_8	V_{CC}				
1 T_C = +25°C	V_{IC}		1 2 3 4 5 6 7 8 9 10 11 12 13 14	-10 mA	-10 mA	-10 mA	-10 mA	-10 mA	-10 mA	-10 mA	-10 mA				GND						-10 mA	-10 mA	-10 mA	-10 mA	-10 mA	-10 mA	4.5 V	A_7 A_6 A_5 A_4 A_3 A_2 A_1 A_0 $\overline{CE}_4$ $\overline{CE}_3$ $\overline{CE}_2$ $\overline{CE}_1$ A_9 A_8		-1.5	V
	V_{OL}	3007	15 16 17 18 19 20 21 22	1/ 2/ 15/	1/	1/	1/	1/	1/	1/17/	1/	18/	18/	18/		18/	18/	18/	18/	18/	5.5 V	5.5 V	0.5 V	0.5 V	1/15/	1/		O_1 O_2 O_3 O_4 O_5 O_6 O_7 O_8		0.5	
	V_{OH}	3006	23 24 25 26 27 28 29 30	1/	1/ 10/	1/9/10/ 12/	1/ 9/	1/ 9/	1/ 9/	1/ 9/	1/9/10/ 19/20/ 12/	-2 mA	-2 mA	-2 mA		-2 mA	-2 mA	-2 mA	-2 mA	-2 mA					1/13/ 19/			O_1 O_2 O_3 O_4 O_5 O_6 O_7 O_8	2.4		
	I_{IL}	3009	31 32 33 34 35 36 37 38 39 40 41 42 43 44	0.5 V	0.5 V	0.5 V	0.5 V	0.5 V	0.5 V	0.5 V	0.5 V										0.5 V	0.5 V	0.5 V	0.5 V	0.5 V	0.5 V	5.5 V	A_7 A_6 A_5 A_4 A_3 A_2 A_1 A_0 $\overline{CE}_4$ $\overline{CE}_3$ $\overline{CE}_2$ $\overline{CE}_1$ A_9 A_8	-1.0	-250	µA
	I_{IH1}	3010	45 46 47 48 49 50 51 52	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V																	A_7 A_6 A_5 A_4 A_3 A_2 A_1 A_0		50	

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TABLE III. Group A inspection for device types 04 and 09 - Continued.

Terminal conditions (outputs not designated are open or resistive coupled to GND or voltage; inputs not designated are high ≥2.0 V, low ≤0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases J,K	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	Measured terminal	Test limits Min	Test limits Max	Unit
			Test no.	A_7	A_6	A_5	A_4	A_3	A_2	A_1	A_0	O_1	O_2	O_3	GND	O_4	O_5	O_6	O_7	O_8	$\overline{CE}_4$	$\overline{CE}_3$	$\overline{CE}_2$	$\overline{CE}_1$	A9	A8	V_{CC}				
1 T_C = +25°C	I_{IH1}	3010	53												GND					5.5 V	5.5 V		5.5 V	5.5 V	5.5 V	5.5 V	$\overline{CE}_4$ $\overline{CE}_3$ $\overline{CE}_1$ A9 A8		50	µA
			54												"					"	"		"	"	"	"			"	"
			55												"					"	"		"	"	"	"			"	"
			56												"					"	"		"	"	"	"			"	"
			57												"					"	"		"	"	"	"			"	"
	I_{IH2} 16/		58												"							4.5 V				"	$\overline{CE}_2$		100	"
	I_{OHZ}		59									5.2 V	5.2 V	5.2 V	"	5.2 V	5.2 V	5.2 V	5.2 V	5.2 V	0.5 V	0.5 V	5.5 V	5.5 V			5.5 V	O_1		"	"
			60												"						"	"	"	"			"	O_2		"	"
			61												"						"	"	"	"			"	O_3		"	"
			62												"						"	"	"	"			"	O_4		"	"
			63												"						"	"	"	"			"	O_5		"	"
			64												"						"	"	"	"			"	O_6		"	"
			65												"						"	"	"	"			"	O_7		"	"
			66												"						"	"	"	"			"	O_8		"	"
	I_{OLZ}		67									0.5 V	0.5 V	0.5 V	"	0.5 V	0.5 V	0.5 V	0.5 V	0.5 V	"	"	"	"			"	O_1	-100	"	"
			68												"						"	"	"	"			"	O_2		"	"
			69												"						"	"	"	"			"	O_3		"	"
			70												"						"	"	"	"			"	O_4		"	"
			71												"						"	"	"	"			"	O_5		"	"
			72												"						"	"	"	"			"	O_6		"	"
			73												"						"	"	"	"			"	O_7		"	"
			74												"						"	"	"	"			"	O_8		"	"
	I_{CC}	3005	75	GND	GND	GND	GND	GND	GND	GND	GND				"						GND	GND	GND	GND	GND	GND	"	V_{CC}		185	mA
	I_{OS}	3011	76	1/	1/10/	1/9/10/	1/ 9/	1/ 9/	1/ 9/	1/ 9/	1/9/10/ 19/20/	GND	GND	GND	"	GND	GND	GND	GND	GND	5.5 V	5.5 V	0.5 V	0.5 V	1/13/ 19/	1/	"	O_1	-15	-100	"
			77	"	"	"	"	"	"	"	"				"						"	"	"	"	"	"	"	O_2	"	"	"
			78	"	"	"	"	"	"	"	"				"						"	"	"	"	"	"	"	O_3	"	"	"
			79	"	"	"	"	"	"	"	"				"						"	"	"	"	"	"	"	O_4	"	"	"
			80	"	"	"	"	"	"	"	"				"						"	"	"	"	"	"	"	O_5	"	"	"
			81	"	"	"	"	"	"	"	"				"						"	"	"	"	"	"	"	O_6	"	"	"
			82	"	"	"	"	"	"	"	"				"						"	"	"	"	"	"	"	O_7	"	"	"
			83	"	"	12/	"	"	"	"	12/				"						"	"	"	"	"	"	"	O_8	"	"	"
2	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = +125°C and V_{IC} tests are omitted.																														
3	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = -55°C and V_{IC} tests are omitted.																														
7 T_C = +25°C	Functional tests	4/	84	4/	4/	4/	4/	4/	4/	4/	4/	4/	4/	4/	GND	4/	4/	4/	4/	4/	4/	4/	4/	4/	4/	4/	4/	Outputs		4/	
8	Same tests, terminal conditions, and limits as for subgroup 7, except T_C = +125°C and T_C = -55°C.																														

See footnotes at end of table.

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TABLE III. Group A inspection for device types 04 and 09 - Continued.

Terminal conditions (outputs not designated are open or resistive coupled to GND or voltage; inputs not designated are high ≥2.0 V, low ≤0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases J,K	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	Measured terminal	Test limits		Unit
			Test no.	A_7	A_6	A_5	A_4	A_3	A_2	A_1	A_0	O_1	O_2	O_3	GND	O_4	O_5	O_6	O_7	O_8	$\overline{CE}_4$	$\overline{CE}_3$	$\overline{CE}_2$	$\overline{CE}_1$	A_9	A_8	V_{CC}		Min	Max	
9 T_C = +25°C	t_{PHL1}	GALPAT Fig. 8	85	5/	5/	5/	5/	5/	5/	5/	5/	6/	6/	6/	GND	6/	6/	6/	6/	6/	5.5 V	5.5 V	GND	GND	5/	5/	5/	Outputs		04 109 90 155	ns
	t_{PLH1}	GALPAT Fig. 8	86	5/	5/	5/	5/	5/	5/	5/	5/	"	"	"	"	"	"	"	"	"	5.5 V	5.5 V	GND	GND	5/	5/	5/	"		90 155	"
	t_{PHL2}	Sequential Fig. 8	87	7/	7/	7/	7/	7/	7/	7/	7/	"	"	"	"	"	"	"	"	"	7/	7/	7/	7/	7/	7/	7/	"		50 130	"
	t_{PLH2}	Sequential Fig. 8	88	7/	7/	7/	7/	7/	7/	7/	7/	"	"	"	"	"	"	"	"	"	7/	7/	7/	7/	7/	7/	7/	"		50 130	"
10	Same tests, terminal conditions, and limits as for subgroup 9, except T_C = +125°C.																														
11	Same tests, terminal conditions, and limits as for subgroup 9, except T_C = -55°C.																														

See footnotes at end of table.

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TABLE III. Group A inspection for device type 05.

Terminal conditions (outputs not designated are open or resistive coupled to GND or voltage; inputs not designated are high ≥2.0 V, low ≤0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases J,K Test no.	1 A_7	2 A_6	3 A_5	4 A_4	5 A_3	6 A_2	7 A_1	8 A_0	9 O_1	10 O_2	11 O_3	12 GND	13 O_4	14 O_5	15 O_6	16 O_7	17 O_8	18 N/C	19 N/C	20 $\overline{CE}$	21 N/C	22 A_9	23 A_8	24 V_{CC}	Measured terminal	Test limits Min	Test limits Max	Unit
1 T_C = +25°C	V_{IC}		1	-10 mA																					4.5 V	A_7		-1.5	V
			2		-10 mA																						A_6			
			3			-10 mA																					A_5			
			4				-10 mA																				A_4			
			5					-10 mA																			A_3			
			6						-10 mA																		A_2			
			7							-10 mA																	A_1			
			8								-10 mA																A_0			
			9																								$\overline{CE}$			
			10																				-10 mA							
			11																						-10 mA		A_9			
																											A_8			
	V_{OL}	3007	12	1/	1/	1/	1/	1/	1/	1/	1/	8 mA	8 mA	8 mA									0.5 V		1/	1/		O_1		0.5	
			13													8 mA											O_2			
			14														8 mA										O_3			
			15															8 mA									O_4			
			16																8 mA								O_5			
			17																	8 mA							O_6			
			18																								O_7			
			19																								O_8			
	V_{OH}	3006	20					1/ 9/	1/ 9/	1/ 9/	1/9/19/	-2 mA	-2 mA	-2 mA											1/9/13/ 19/			O_1	2.4		
			21													-2 mA											O_2			
			22														-2 mA										O_3			
			23															-2 mA									O_4			
			24																-2 mA								O_5			
			25																								O_6			
			26																								O_7			
			27																								O_8			
	I_{IL}	3009	28	0.5 V																					5.5 V	A_7	-1	-250	µA
			29		0.5 V																					A_6			
			30			0.5 V																				A_5			
			31				0.5 V																			A_4			
			32					0.5 V																		A_3			
			33						0.5 V																	A_2			
			34							0.5 V																A_1			
			35								0.5 V															A_0			
			36																				0.5 V			$\overline{CE}$			
			37																						0.5 V	A_9			
			38																							A_8			
	I_{IH}	3010	39	5.5 V																					5.5 V	A_7		50	µA
			40		5.5 V																					A_6			
			41			5.5 V																				A_5			
			42				5.5 V																			A_4			
			43					5.5 V																		A_3			
			44						5.5 V																	A_2			
			45							5.5 V																A_1			
			46								5.5 V															A_0			
			47																				5.5 V			$\overline{CE}$			
			48																						5.5 V	A_9			
			49																							A_8			

See footnotes at end of table.

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TABLE III. Group A inspection for device type 05 - Continued.

Terminal conditions (outputs not designated are open or resistive coupled to GND or voltage; inputs not designated are high ≥2.0 V, low ≤0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases J, K Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	Measured terminal	Test limits Min	Test limits Max	Unit
				A_7	A_6	A_5	A_4	A_3	A_2	A_1	A_0	O_1	O_2	O_3	GND	O_4	O_5	O_6	O_7	O_8	N/C	N/C	$\overline{CE}$	N/C	A_9	A_8	V_{CC}				
1 T_C = +25°C	I_{OHZ}		50 51 52 53 54 55 56 57									5.2 V	5.2 V	5.2 V	GND " " " " " " "	5.2 V	5.2 V	5.2 V	5.2 V	5.2 V			5.5 V " " " " " " "				5.5 V " " " " " " "	O_1 O_2 O_3 O_4 O_5 O_6 O_7 O_8		100 " " " " " " "	µA " " " " " " "
	I_{OLZ}		58 59 60 61 62 63 64 65									0.5 V	0.5 V	0.5 V	" " " " " " " "	0.5 V	0.5 V	0.5 V	0.5 V	0.5 V			" " " " " " " "				" " " " " " " "	O_1 O_2 O_3 O_4 O_5 O_6 O_7 O_8		-100 " " " " " " "	" " " " " " " "
	I_{OS}	3011	66 67 68 69 70 71 72 73	1/ " " " " " " "	1/ " " " " " " "	1/ " " " " " " "	1/ " " " " " " "	1/ 9/ " " " " " " "	1/ 9/ " " " " " " "	1/ 9/ " " " " " " "	1/9/19/ " " " " " " "	GND	GND	GND	" " " " " " " "	GND	GND	GND	GND	GND			0.5 V " " " " " " "		1/9/13/ 19/ " " " " " "	1/ " " " " " " "	" " " " " " " "	O_1 O_2 O_3 O_4 O_5 O_6 O_7 O_8	-15 " " " " " " "	" " " " " " " "	" " " " " " " "
	I_{CC}	3005	74	GND	GND	GND	GND	GND	GND	GND	GND				"								GND		GND	GND	"	V_{CC}		185	mA
2	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = +125°C and V_{IC} tests are omitted.																														
3	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = -55°C and V_{IC} tests are omitted.																														
7 T_C = +25°C	Functional tests	4/	75	4/	4/	4/	4/	4/	4/	4/	4/	4/	4/	4/	GND	4/	4/	4/	4/	4/			GND		4/	4/	4/	Outputs		4/	
8	Same tests, terminal conditions, and limits as for subgroup 7, except T_C = +125°C and T_C = -55°C.																														
9 T_C = +25°C	t_{PHL1}	GALPAT Fig. 8	76	5/	5/	5/	5/	5/	5/	5/	5/	6/	6/	6/	GND	6/	6/	6/	6/	6/			GND		5/	5/	5/	Outputs		8/	ns
	t_{PLH1}	GALPAT Fig. 8	77	5/	5/	5/	5/	5/	5/	5/	5/	"	"	"	"	"	"	"	"	"			GND		5/	5/	5/	"		"	"
	t_{PHL2}	Sequential Fig. 8	78	7/	7/	7/	7/	7/	7/	7/	7/	"	"	"	"	"	"	"	"	"			7/		7/	7/	7/	"		"	"
	t_{PLH2}	Sequential Fig. 8	79	7/	7/	7/	7/	7/	7/	7/	7/	"	"	"	"	"	"	"	"	"			7/		7/	7/	7/	"		"	"
10	Same tests, terminal conditions, and limits as for subgroup 9, except T_C = +125°C.																														
11	Same tests, terminal conditions, and limits as for subgroup 9, except T_C = -55°C.																														

See footnotes at end of table.

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TABLE III. Group A inspection for device type 06.

Terminal conditions (outputs not designated are open or resistive coupled to GND or voltage; inputs not designated are high ≥2.0 V, low ≤0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases J,K	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	Measured terminal	Test limits Min	Test limits Max	Unit
			Test no.	A_7	A_6	A_5	A_4	A_3	A_2	A_1	A_0	O_1	O_2	O_3	GND	O_4	O_5	O_6	O_7	O_8	N/C	N/C	$\overline{CE}$	N/C	A_9	A_8	V_{CC}				
1 T_C = +25°C	V_{IC}		1	-10 mA																						4.5 V	A_7		-1.5	V
			2		-10 mA																							A_6			
			3			-10 mA																						A_5			
			4				-10 mA																					A_4			
			5					-10 mA																				A_3			
			6						-10 mA																			A_2			
			7							-10 mA																		A_1			
			8								-10 mA																	A_0			
			9																				-10 mA					$\overline{CE}$			
			10																						-10 mA			A_9			
			11																							-10 mA		A_8			
	V_{OL}	3007	12	1/	1/	1/	1/	1/	1/	1/	1/	8 mA											0.5 V		1/	1/		O_1		0.5	
			13	"	"	"	"	"	"	"	"		8 mA			8 mA							"		"	"		O_2			
			14	"	"	"	"	"	"	"	"			8 mA			8 mA						"		"	"		O_3			
			15	"	"	"	"	"	"	"	"				8 mA			8 mA					"		"	"		O_4			
			16	"	"	"	"	"	"	"	"					8 mA			8 mA				"		"	"		O_5			
			17	"	"	"	"	"	"	"	"						8 mA			8 mA			"		"	"		O_6			
			18	"	"	"	"	"	"	"	"							8 mA					"		"	"		O_7			
			19	"	"	"	"	"	"	"	"								8 mA				"		"	"		O_8			
	I_{IL}	3009	20	0.5 V																						5.5 V	A_7	-1.0	-250	μA
			21		0.5 V																		0.5 V			"	A_6	"	"	
			22			0.5 V																	"				"	A_5	"	"	
			23				0.5 V																"				"	A_4	"	"	
			24					0.5 V															"				"	A_3	"	"	
			25						0.5 V														"				"	A_2	"	"	
			26							0.5 V													"				"	A_1	"	"	
			27								0.5 V												"				"	A_0	"	"	
			28																				0.5 V					$\overline{CE}$			
			29																						0.5 V			A_9			
			30																							0.5 V		A_8			
	I_{IH1}	3010	31	5.5 V																						5.5 V	A_7		50	μA
			32		5.5 V																						"	A_6	"	"	
			33			5.5 V																					"	A_5	"	"	
			34				5.5 V																				"	A_4	"	"	
			35					5.5 V																			"	A_3	"	"	
			36						5.5 V																		"	A_2	"	"	
			37							5.5 V																	"	A_1	"	"	
			38								5.5 V																"	A_0	"	"	
			39																						0.5 V			A_9			
			40																							0.5 V		A_8			
	I_{IH2}		41																				4.5 V				"	$\overline{CE}$		100	"
	I_{CEX}		42									5.2 V														5.5 V	O_1			
			43										5.2 V														"	O_2			
			44											5.2 V													"	O_3			
			45													5.2 V											"	O_4			
			46														5.2 V										"	O_5			
			47															5.2 V									"	O_6			
			48																5.2 V								"	O_7			
			49																	5.2 V							"	O_8			

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TABLE III. Group A inspection for device type 06 - Continued.

Terminal conditions (outputs not designated are open or resistive coupled to GND or voltage; inputs not designated are high ≥2.0 V, low ≤0.8 V, or open).

Subgroup	Symbol	MIL-STD-883 method	Cases J,K	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	Measured terminal	Test limits		Unit
			Test no.	A_7	A_6	A_5	A_4	A_3	A_2	A_1	A_0	O_1	O_2	O_3	GND	O_4	O_5	O_6	O_7	O_8	N/C	N/C	CE	N/C	A_9	A_{10}	V_{CC}		Min	Max	
1 T_C = +25°C	I_{CC}	3005	50	GND	GND	GND	GND	GND	GND	GND	GND				GND								GND		GND	GND	5.5 V	V_{CC}		185	mA
2	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = +125°C and V_{IC} tests are omitted.																														
3	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = -55°C and V_{IC} tests are omitted.																														
7 T_C = +25°C	Functional tests	4/	51	4/	4/	4/	4/	4/	4/	4/	4/	4/	4/	4/	GND	4/	4/	4/	4/	4/			4/			4/	4/	Outputs		4/	
8	Same tests, terminal conditions, and limits as for subgroup 7, except T_C = +125°C and T_C = -55°C.																														
9 T_C = 25°C	t_{PHL1}	GALPAT Fig. 8	52	5/	5/	5/	5/	5/	5/	5/	5/	6/	6/	6/	GND	6/	6/	6/	6/	6/			GND		5/	5/	5/	Outputs		8/	ns
	t_{PLH1}	GALPAT Fig. 8	53	5/	5/	5/	5/	5/	5/	5/	5/	"	"	"	"	"	"	"	"	"			GND		5/	5/	5/	"		"	"
	t_{PHL2}	Sequential Fig. 8	54	7/	7/	7/	7/	7/	7/	7/	7/	"	"	"	"	"	"	"	"	"			7/		7/	7/	7/	"		"	"
	t_{PLH2}	Sequential Fig. 8	55	7/	7/	7/	7/	7/	7/	7/	7/	"	"	"	"	"	"	"	"	"			7/		7/	7/	7/	"		"	"
10	Same tests, terminal conditions, and limits as for subgroup 9, except T_C = +125°C.																														
11	Same tests, terminal conditions, and limits as for subgroup 9, except T_C = -55°C.																														

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1/ For programmed devices, select an appropriate address to acquire the desired output state.

2/ For unprogrammed device types 01 and 02 (circuit A), apply 10.0 V on pin 1 (A_6); and for unprogrammed device types 03 and 04, apply 10.0 V on pin 1 (A_7) (circuit A).

3/ I_{OL} = 12 mA for circuits A, C, E, and G devices; I_{OL} = 16 mA for circuits B, D, and F devices.

4/ The functional tests shall verify that no fuses are blown for unprogrammed devices or that the truth table specified in the altered item drawing exists for programmed devices (see table II and 3.2.2.2). All bits shall be tested. The functional tests shall be performed with V_{CC} = 4.5 V and V_{CC} = 5.5 V. Terminal conditions shall be as follows:

a. Inputs: H = 3.0 V, L = 0.0 V.
b. Outputs: Output voltage shall be either:
   (1) H = 2.4 V minimum and L = 0.5 V maximum when using a high-speed checker double comparator, or
   (2) H $\geq$1.0 V and L <1.0 V when using a high-speed checker single comparator.

5/ GALPAT (PROGRAMMED PROM). This program will test all bits in the array, the addressing and interaction between bits for ac performance, t_{PHL1} and t_{PLH1}. Each bit in the pattern is fixed by being programmed with an "H" or "L". The GALPAT tests shall be performed with V_{CC} = 4.5 V and V_{CC} = 5.5 V.

Description:

Step 1. Word 0 is read.
Step 2. Word 1 is read.
Step 3. Word 0 is read.
Step 4. Word 2 is read.
Step 5. Word 0 is read.
Step 6. The reading procedure continues back and forth between word 0 and the next higher numbered word until word 1023 or 2047 (as applicable) is reached, then increments to the next word and reads back and forth as in Step 1 through Step 6 and shall include all words.
Step 7. Pass execution time = $(n^2 + n)$ x cycle time. n = 1024 or 2048 (as applicable).

6/ The outputs are loaded per figure 8.

7/ SEQUENTIAL (PROGRAMMED PROM). This program will test all bits in the array for t_{PHL2} and t_{PLH2}. The sequential tests shall be performed with V_{CC} = 4.5 V and V_{CC} = 5.5 V.

Description:

Step 1. Each word in the pattern is tested from the enable lines to the output lines for recovery.
Step 2. Word 0 is addressed. Enable line is pulled high to low and low to high. t_{PHL2} and t_{PLH2} are read.
Step 3. Word 1 is addressed. Same enable sequence as above.
Step 4. The reading procedure continues until word 1023 or 2047 (as applicable) is reached.
Step 5. Pass execution times 1024 or 2048 (as applicable) x cycle time.

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8/

Device type	t_{PHL1} (ns)	t_{PLH1} (ns)	t_{PHL2} (ns)	t_{PLH2} (ns)
01, 02	125	125	60	60
03, 04, 05, 06	90	90	50	50
Circuit F	90	90	50	50
Circuit B device 08	55	55	30	30
09	55	55	30	30
10	55	55	30	30

9/ For unprogrammed devices (circuit C), apply 10.0 V on pin 15 (A_9), 0.5 V on pin 2 (A_5) and 5.0 V to all other address pins for device types 02 and 10; device types 04 and 09, apply 10 V on pin 8 (A_0) and 5.0 V on pins 7, 6, 5, 4, 3 (A_1, A_2, A_3, A_4, A_5); device type 05, apply 10 V on pin 22 (A_9), 0.5 V on pins 8, 7, 6, 5 (A_0, A_1, A_2, A_3) and 5.0 V to all other address pins. For unprogrammed devices (circuit F) apply 12.0 V on pin 5 (A_0) for device types 02 and 08.

10/ For unprogammed devices (circuit G), apply 10.5 V to pins 1 and 8 (A_6 and A_{10}), apply 0.0 V to pin 2 (A_5) and apply 3.0 V to all other address pins for device types 01 and 02; apply 10.5 V to pin 3 (A_5), apply 0.0 V to pins 2 and 8 (A_6 and A_0), and apply 3.0 V to all other address pins for device types 03 and 04.

11/ For unprogrammed devices, apply 12.0 V on pin 1 (A_6) for device types 02 and 08 (circuit B).

12/ For unprogrammed devices (circuit G), apply 10.5 V on pins 1 and 8 (A_6 and A_{10}), apply 0.0 V to pin 3 (A_4) and apply 3.0 V to all other address pins for device type 02, apply 10.5 V to pin 3 (A_5), apply 0.0 V to pin 8 (A_0) and apply 3.0 V to all other address pins for device type 04.

13/ For unprogrammed device type 02 (with date codes before 8501), apply 10.0 V on pin 5 (A_0); 0.5 V on pin 16 (A_8), and 5.0 V on all other address pins; and for unprogrammed device type 04 (with date codes before 8501) (circuit C), apply 10.0 V on pin 22 (A_9) and 5.0 V on all other address pins.

14/ For circuit B, device type 08, apply 2.4 V.

15/ For unprogrammed devices (circuit B), apply 12.0 V on pins 22 and 1 (A_9 and A_7) for device types 03 and 04.

16/ At the manufacturer's option, this may be performed with V_{IN} = 5.5 V and test limits of 50 µA maximum.

17/ For unprogrammed devices (circuit F) apply 12.0 V on pin 6 (A_2) and all other inputs at 0 V for device type 04.

18/ I_{OL} = 8 mA for circuits A, B, C, D, E, and G devices; I_{OL} = 16 mA for circuit F devices.

19/ For unprogrammed devices (circuit D), apply 12.0 V on pins 8 and 22 (A_0 and A_9), select an appropriate address to acquire the desired output state.

20/ For unprogrammed device type 03 (circuit E), apply 13.0 V on pin 4 (A_4) and pin 8 (A_0); and for unprogrammed device type 04 (circuit E), apply 13.0 V on pin 8 (A_0).

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4.7 Programming procedures for circuit A. The programming characteristics of table IVA and the following procedures shall be used for programming the device:

a. Connect the device in the electrical configuration for programming. The waveforms on figure 6A and the programming characteristics of table IVA shall apply to these procedures.

b. Address the PROM with the binary address of the word to be programmed. Address inputs are TTL compatible. An open circuit shall not be used to address the PROM.

c. Apply V_{PL} voltage to V_{CC}.

d. Bring the $\overline{CE}_x$ inputs high and the CE_x inputs low to disable the device. The chip enables are TTL compatible. An open circuit shall not be used to disable the device.

e. Disable the programming circuitry by applying a voltage of V_{OPD} to the outputs of the PROM.

f. Raise V_{CC} to V_{PH} with rise time less than or equal to t_{TLH}.

g. After a delay equal to or greater than t_{D1}, apply only one pulse with amplitude of V_{OPE} and duration of t_p to the output selected for programming. Note that the PROM is supplied with fuses intact, which generates an output high. Programming a fuse will cause the output to go low.

h. Lower V_{CC} to V_{PL} following a delay of t_{D2} from programming enable pulse applied to an output.

i. Enable the PROM for verification by applying V_{IL} to $\overline{CE}_x$ and V_{IH} to CE_x.

j. Apply V_{PHV} to V_{CC} and verify bit is programmed.

k. Repeat 4.7a through 4.7j for all other bits to be programmed in the PROM.

l. For classes S and B devices, if any bit does not verify as programmed, it shall be considered a programming reject.

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TABLE IVA. Programming characteristics for circuit A. 1/

Characteristic	Symbol	Limits 2/			Unit
		Min	Recommended	Max	
Address input voltage 3/	V_{IH}	2.4	5.0	5.0	V
	V_{IL}	0.0	0.4	0.5	V
Programming	V_{PH} 4/	10.75	11.0	11.25	V
Voltage to V_{CC} low	V_{PL}	0.0	0.0	1.5	V
Program verify	V_{PHV}		5.5		V
Verify voltage	V_R 5/	4.5		5.5	V
Programming input low current at V_{PH}	I_{ILP}		-300	-600	µA
Programming voltage (V_{CC}) transition time	t_{TLH}	1	5	10	µs
	t_{THL}	1	5	10	µs
Programming delay	t_{D1}	10	10	20	µs
	t_{D2}	1	5	5	µs
Programming pulse width	t_p 6/	90	100	110	µs
Programming duty cycle	PDC		30	60	%
Output voltage, enable	V_{OPE} 7/	10.5	10.5	11.0	V
Output voltage, disable	V_{OPD}	0.0	5.0	5.5	V

1/ During the programming, the chip must be disabled for proper operation.

2/ T_C = +25°C.

3/ No inputs should be left open for V_{IH}.

4/ V_{PH} source must be capable of supplying one ampere.

5/ It is recommended that post programming dual verification be made at V_R min and V_R max.

6/ Note 4.7j in programming procedure.

7/ V_{OPE} source must be capable of supplying 10 mA minimum.

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4.8 Programming procedures for circuit B, device types 03 and 04. The waveforms on figure 6B, the programming characteristics of table IVB, and the following procedures shall apply:

a. Connect the device in the electrical configuration for programming.

b. Raise V_{CC} to 5.5 volts.

c. Address the PROM with the binary address of the selected word to be programmed. Address inputs are TTL compatible.

d. Disable the chip by applying V_{IH} to the CE inputs and V_{IL} to the CE inputs. The chip enable inputs are TTL compatible.

e. Apply the V_{PP} pulse to the programming pin (CE_2). In order to insure that the output transistor is off before increasing the voltage on the output pin, the programming pin's voltage pulse shall precede the output pin's programming pulse by T_{D1} and leave after the output pin's programming pulse by T_{D2} (see figure 6B).

f. Apply only one V_{OUT} pulse with duration of t_p to the output selected for programming. The outputs shall be programmed one output at a time, since internal decoding circuitry is capable of sinking only one unit of programming current at a time. Note that the PROM is supplied with fuses generating a high-level logic output. Programming a fuse will cause the output to go to a low-level logic in the verify mode.

g. Other bits in the same word may be programmed sequentially by applying V_{OUT} pulses to each output to be programmed.

h. Repeat 4.8c through 4.8g for all other bits to be programmed.

i. Enable the chip by applying V_{IL} to the $\overline{CE}$ inputs and V_{IH} to the CE inputs, and verify the program. Verification may check for a low output by requiring the device to sink 12 mA at V_{CC} = 4.0 V and 0.2 mA at V_{CC} = 7.0 V at T_C = +25°C.

j. For classes S and B devices, if any bit does not verify as programmed, it shall be considered a programming reject.

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TABLE IVB. Programming characteristics for circuit B, device types 03 and 04.

Characteristic	Symbol	Conditions	Limits 1/			Unit
			Min	Recommended	Max	
V_{CC} required during programming	V_{CCP}		5.4	5.5	5.6	V
Rise time of programming pulse to data out or programming pin	t_{TLH}		0.34	0.40	0.46	V/µs
Programming voltage on programming pin	V_{PP}		32.5	33	33.5	V
Output programming voltage	V_{OUT}		25.5	26	26.5	V
Programming pin pulse width ($\overline{CE}_2$)	t_{PP}	Chip disabled, V_{CC} = 5.5 V		100	180	ns
Pulse width of programming voltage	t_P	Chip disabled, V_{CC} = 5.5 V	1		40	µs
Required current limit of power supply feeding programming pin and output during programming	I_L	V_{PP} = 33 V, V_{OUT} = 26 V, V_{CC} = 5.5 V	240			mA
Required time delay between disabling memory output and application of output programming pulse	T_{D1}	Measured at 10% levels	70	80	90	µs
Required time delay between removal of programming pulse and enabling memory output	T_{D2}	Measured at 10% levels	100			ns
Output current during verification	I_{OLV1}	Chip enabled, V_{CC} = 4.0 V	11	12	13	mA
	I_{OLV2}	Chip enabled, V_{CC} = 7.0 V	0.19	0.2	0.21	mA
Address input voltage	V_{IH}		2.4	5.0	5.5	V
	V_{IL}		0.0	0.4	0.8	V
Maximum duty cycle during automatic programming of programming pin and output pin	D.C.	t_P/t_C			25	%

1/ T_C = +25°C.

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4.8.1 Programming procedures for circuit B, device types 01, 02, and 08. The waveforms on figure 6B (device types 01, 02, and 08), the programming characteristics of table IVB (device types 01, 02, and 08), and the following shall apply:

a. Connect the device in the electrical configuration for programming.

b. Apply V_{IH} to $\overline{CE}_1$ and the binary address of the PROM word to be programmed. Raise V_{CC} to V_{CCP}.

c. After a t_D delay, apply only one V_{OP} to the output to be programmed high. Apply V_{OP} to one output at a time.

d. After a t_D delay, a pulse $\overline{CE}_1$ to a V_{IL} level for a duration of t_P.

e. After a t_P and a t_D delay, remove V_{OP} from the programmed output.

f. Other bits in the same word may be programmed sequentially while the V_{CC} input is at the V_{CCP} level by applying V_{OP} pulses to each output to be programmed and pulsing $\overline{CE}_1$ to the V_{IL} level, allowing for proper delays between V_{OP} and $\overline{CE}_1$.

g. Repeat 4.8.1b through 4.8.1e for all bits to be programmed.

h. To verify programming, lower V_{CCP} to V_{CC}. Connect a 10-kΩ resistor between each output and V_{CC}. Apply V_{IL} to $\overline{CE}_1$ input. The programmed outputs should remain in the high state and the unprogrammed outputs should go to the low level.

i. For classes S and B devices, if any bit does not verify as programmed, it shall be considered a programming reject.

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TABLE IVB. Programming characteristics for circuit B, device types 01, 02, and 08.

Characteristic	Symbol	Conditions	Limits 1/			Unit
			Min	Recommended	Max	
V_{CC} required during programming	V_{CCP}		11.5	11.75	12.0	V
V_{OUT} current limit during programming	I_{OP}		20	25	30	mA
Output programming voltage	V_{OUT}		10.5	11.0	11.5	V
Pulse width of programming voltage	t_P		9	10	11	µs
Programming delay	t_D		0	1	10	µs
V_{CCP} or V_{OUT} transition time	t_{TLH}	Rise time of V_{CC} or V_{OUT}	1	5	10	V/µs
V_{CCP} current	I_{CCP}		800	900	1000	mA
Low V_{CC} for verification	V_{CCL}		4.2	4.3	4.4	V
High V_{CC} for verification	V_{CCH}		5.8	6.0	6.2	V
Address input voltage	V_{IH}		2.4	3.0	5.5	V
	V_{IL}		0.0	0.0	0.5	V
Maximum duty cycle during automatic programming of programming pin and output pin	D.C.	t_P/t_C		25	25	%

1/ T_C = +25°C.

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4.9 Programming procedures for circuit C. The waveforms on figure 6C, the programming characteristics of table IVC, and the following procedures shall apply:

a. Connect the device in the electrical configuration for programming.

b. Terminate all device outputs with a 10-kΩ resistor to V_{CC}. Apply V_{IH} to the $\overline{CE}$ inputs and V_{IL} to the CE inputs.

c. Address the PROM with the binary address of the selected word to be programmed. Raise V_{CC} to V_{CCP}.

d. After a t_D delay (10 µs), apply only one V_{OUT} pulse to the output to be programmed. Program one output at a time.

e. After a t_D delay (10 µs), pulse $\overline{CE}$ input to logic "0" for a duration of t_p.

f. After a t_D delay (10 µs), remove the V_{OUT} pulse from the programmed output. Programming a fuse will cause the output to go to a high-level logic in the verify mode.

g. Other bits in the same word may be programmed sequentially while the V_{CC} input is at the V_{CCP} level by applying V_{OUT} pulses to each output to be programmed allowing a delay of t_D between pulses as shown on figure 6C.

h. Repeat 4.9c through 4.9g for all other bits to be programmed.

i. To verify programming after t_D (10 µs) delay, lower V_{CC} to V_{CCH} and apply a logic "0" level to $\overline{CE}$ inputs and logic "1" level to CE inputs. The programmed output should remain in the "1" state. Again, lower V_{CC} to V_{CCL} and verify that the programmed output remains in the "1" state.

j. For classes S and B devices, if any bit does not verify as programmed, it shall be considered a programming reject.

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TABLE IVC. Programming characteristics for circuit C.

Characteristic	Symbol	Conditions	Limits			Unit
			Min	Recommended	Max	
Programming voltage	V_{CCP} 1/	I_{CCP} = 375 ±75 mA, transient or steady state	8.5	8.75	9.0	V
Verification upper limit	V_{CCH}		5.3	5.5	5.7	V
Verification lower limit	V_{CCL}		4.3	4.5	4.7	V
Verify threshold	V_S 2/		1.4	1.5	1.6	V
Programming supply current	I_{CCP}	V_{CCP} = 8.75 ±.25 V	300	350	400	mA
Input voltage high level "1"	V_{IH}		2.4		5.5	V
Input voltage low level "0"	V_{IL}		0	0.4	0.8	V
Input current	I_{IH}	V_{IH} = 5.5 V			50	µA
Input current	I_{IL}	V_{IL} = 0.4 V			-500	µA
Output programming voltage	V_{OUT} 3/	I_{OUT} = 200 ±20 mA, transient or steady state	16	17	18	V
Output programming current	I_{OUT}	V_{OUT} = 17 ±1 V	180	200	220	mA
Programming voltage transition time	t_{TLH}		10		50	µs
$\overline{CE}$ programming pulse width	t_P		0.3	0.4	0.5	ms
Pulse sequence delay	t_D		10			µs
Programming duty cycle	t_{PR} / $t_{PR}+t_{PS}$				50	%

1/ Bypass V_{CC} to GND with a 0.01-µF capacitor to reduce voltage spikes.

2/ V_S is the sensing threshold of the PROM output voltage for a programmed bit. It normally constitutes the reference voltage applied to a comparator circuit to verify a successful fusing attempt.

3/ Care should be taken to insure the 17 ±1 V output voltage is maintained during the entire fusing cycle. The recommended supply is a constant current source clamped at the specified voltage limit.

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4.10 Programming procedures for circuit D. The programming characteristics on table IVD and the following procedures shall be used for programming the device:

a. Connect the device in the electrical configuration for programming. The waveforms on figure 6D and the programming characteristics of table IVD shall apply to these procedures.

b. Select the word to be programmed by applying the appropriate voltages to the address pins as well as the required voltages to chip enable pins to select the device.

c. Apply the proper power, V_{CC} = 6.5 V, GND = 0 V.

d. Verify that the bit to be programmed is in the "0" logic state.

e. Enable the chip for programming by application of the chip enable voltage, $V_{P(CE)}$ = 21.0 V, $\overline{CE}_2$, CE_3 and CE_4 should be left high, and $\overline{CE}_1$ should remain low.

f. Apply I_{OP} programming current ramp to the output to be programmed. The other outputs shall be left open. Only one output may be programmed at a time. During the rise of the current ramp, the required current will be achieved to program the junction. As programming occurs, a drop in voltage can be sensed at the output of the device. Upon detection of V_{PS}, the current shall be held for t_{hap} and then shut off.

g. Verify that the programmed bit is in the "1" logic state. Lower $V_{P(CE_1)}$ to 0 V and read the output. Note that the PROM is supplied with fuses generating a low-level logic output. Programming a fuse will cause the output to go to a high-level logic in the verify mode.

h. Lower V_{CC} to 0 V. The power supply duty cycle shall be equal to or less than 50 percent.

i. If the bit verifies as not having been programmed at V_{CC} = 6.5 V, repeat the programming ramp sequence up to 15 times until the bit is programmed. If after 16 programming attempts, the bit does not program, the device shall be considered a reject.

j. If the bit verifies as having been programmed at V_{CC} = 6.5 V, one of the following two conditions shall be followed:

(1) If the current required to program was less than I_{OP}(max), proceed to 4.10 k.

(2) If the current required to program was equal to or greater than I_{OP}(max), the device shall be considered a reject and no further attempts at programming other bits shall be attempted.

k. Repeat 4.10a through 4.10j for all other bits to be programmed.

l. For classes S and B devices, if any bit does not verify as programmed, it shall be considered a programming reject.

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TABLE IVD. Programming characteristics for circuit D.

Characteristic	Symbol	Conditions 1/	Limits			Unit
			Min	Recommended	Max	
Address input voltage	V_{IH}	Don't leave inputs open	2.4	5.0	5.0	V
	V_{IL}		0	0	0.4	
Chip enable programming voltage	$V_{P(CE)}$	$\overline{CE_1} = V_{IL}$ $CE_3 = \overline{CE_4} = V_{IH}$ $V_{P(CE)} = \overline{CE_2}$	20.5	21.0	21.5	V
Programming voltage limit	$V_{OP(max)}$	Programming current ramp voltage limit	24	25	26	V
Power supply	V_{CC}		6.3	6.5	6.7	V
Power supply current	I_{CC}				250	mA
Chip enable current	I_{CE}				150	mA
Initial value of programming current ramp	$I_{OP(INIT)}$		19	20	21	mA
Maximum value of programming current ramp	$I_{OP(max)}$		155	160	165	mA
Programming current ramp (linear slew rate)	SRI_{OP}		0.9	1.0	1.1	mA/µs
V_{CC} pulse rise time	$t_r(V_{CC})$		0.2	2.0		µs
V_{CC} pulse fall time	$t_f(V_{CC})$		0.2	2.0		µs
Chip enable rise time	$t_r(\overline{CE_2})$		3.0	4.0		µs
Chip enable fall time	$t_f(\overline{CE_2})$		0.2	4.0		µs
Programming current ramp fall time	$t_f(I_{OP})$			0.1	0.2	µs
Hold time after programming	t_{hap}		1.4	1.5	1.6	µs
Time to reach I_{OP} initial	t_{IOP}		0.5	1.0	2.0	µs
Delay to start V_{ps} sense	t_{dss}		2.0	3.0	4.0	µs

See footnote at end of table.

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TABLE IVD. Programming characteristics for circuit D - Continued.

Characteristic	Symbol	Conditions 1/	Limits			Unit
			Min	Recommended	Max	
Delay to chip enable pulse	t_{dce}			1.0		µs
Delay to programming ramp	$t_{d}(I_{OP})$		2.0	3.0	10	µs
Delay after programming to CE_1	t_{dRAP}		2.0	3.0	10	µs
Delay to read after programming	t_{dRAP}	Programming verification	2.0	3.0		µs
Delay to V_{CC} off	$t_{D}(V_{CC})$			1.0		µs
Delay to read before programming	t_{dRBP}	Initial check	2.0	3.0		µs
Width to read compare strobe	t_{w}			1.0		µs
Voltage change at programming	V_{ps}	Typical 2.0 V	0.7	2.0		V
Time to program bit	t_{tp}	V_{ps} sensing circuit will automatically adjust this time				
Duty cycle power		Maximum duty cycle to maintain T_C < +85°C		50		%
Case temperature	T_C		25	85		°C

1/ T_C = +25°C.

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4.11 Programming procedures for circuit E. The waveforms on figure 6E, the programming characteristics of table IVE, and the following procedures shall apply:

a. Connect the device in the electrical configuration for programming.

b. Terminate all outputs with a 300-ohm resistor to V_{ONP}. Apply V_{IHP} to the $\overline{CE}_2$, CE_3, and CE_4 inputs and V_{ILP} to the $\overline{CE}_1$ inputs.

c. Address the PROM with the binary address of the selected word to be programmed. Raise V_{CC} to V_{CCP}.

d. After a delay of t_1, apply only one V_{OP} pulse with a duration of t_p, t_2, and $d(V_{OP})/dt$ to the output selected for programming. After a delay of t_2 and $d(V_{OP})/dt$, pulse $\overline{CE}_2$ from V_{IHP} to V_{CEP} for the duration of t_p, $2d(V_{CE})/dt$, and t_3; $\overline{CE}_2$ is then to go to the V_{ILP} level.

e. To verify programming after $\overline{CE}_1$ has been set to V_{ILP}, lower V_{CC} to V_{CCL} after a delay of t_4. The programmed output should remain in the logic "1" state.

f. The outputs should be programmed one output at a time since the internal decoding circuitry is capable of sinking only one unit of programming current at a time. Note that the PROM is supplied with fuses generating a low-level logic output. Programming a fuse will cause the output to go to a high-level logic in the verify mode.

g. Repeat 4.11c through 4.11f for all other bits to be programmed.

h. For classes S and B devices, if any bit does not verify as programmed, it shall be considered a programming reject.

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TABLE IVE. Programming characteristics for circuit E.

Characteristic	Symbol	Conditions	Limits			Unit
			Min	Recommended	Max	
V_{CC} required during programming	V_{CCP}		5.0		5.5	V
High-level input voltage during programming	V_{IHP}		2.4		5.5	V
Low-level input voltage during programming	V_{ILP}		0.0		0.45	V
Chip enable voltage during programming	V_{CEP}	$\overline{CE}_1$ pin	14.5		15.5	V
Output voltage during programming	V_{OP}		19.5		20.5	V
Voltage on outputs not to be programmed	V_{ONP}		0		V_{CCP} +0.3	V
Current on outputs not to be programmed	I_{ONP}				20	mA
Rate of output voltage change	$\frac{d(V_{OP})}{dt}$		20		250	V/µs
Rate of chip enable voltage change	$\frac{d(V_{CE})}{dt}$	$\overline{CE}_1$ pin	100		1000	V/µs
Programming period	tp		50		100	µs
V_{CC} during programming verification	V_{CCL}		4.5		5.0	µs

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4.12 Programming procedures for circuit F. The waveforms on figure 6F, the programming characteristics of table IVF, and the following procedures shall apply:

a. Connect the device in the electrical configuration for programming.

b. Raise V_{CC} to 5.5 volts.

c. Address the PROM with the binary address of the selected word to be programmed. Address inputs are TTL compatible.

d. Disable the chip by applying V_{IH} to the $\overline{CE}$ inputs and V_{IL} to the $\overline{CE}$ inputs. The chip enable inputs are TTL compatible.

e. Apply the V_{PP} pulse to the programming pin ($\overline{CE}_2$). In order to insure that the output transistor is off before increasing voltage on the output pin, the programming pin's voltage pulse shall precede the output pin's programming pulse by T_{D1} and leave after the programming pin's programming pulse by T_{D2} (see figure 6F).

f. Apply only one V_{OUT} pulse with duration of t_p to the output selected for programming. The outputs shall be programmed one output at a time since internal decoding circuitry is capable of sinking only one unit of programming current at a time. Note that the PROM is supplied with fuses generating a high-level logic output. Programming a fuse will cause the output to go to a low-level logic in the verify mode.

g. Other bits in the same word may be programmed sequentially by applying V_{OUT} pulses to each output to be programmed.

h. Repeat 4.12c through 4.12g for all other bits to be programmed.

i. Enable the chip by applying V_{IL} to the $\overline{CE}$ inputs and V_{IH} to the $\overline{CE}$ inputs, and verify the program. Verification may check for a low output by requiring the device to sink 12 mA at V_{CC} = 4.2 V and 0.2 mA at V_{CC} = 6.2 V at T_C = +25°C.

j. For classes S and B devices, if any bit does not verify as programmed, it shall be considered a programming reject.

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TABLE IVF. Programming characteristics for circuit F.

Characteristic	Symbol	Conditions	Limits 1/			Unit
			Min	Recommended	Max	
V_{CC} required during programming	V_{CCP}		5.4	5.5	5.6	V
Rise time of programming pulse to data out or programming pin	t_{TLH}		0.34	0.40	0.46	V/µs
Programming voltage on programming pin	V_{PP}		32.5	33	33.5	V
Output programming voltage	V_{OUT}		25.5	26	26.5	V
Programming pin pulse width ($\overline{CE}$)	t_{PP}	Chip disabled, V_{CC} = 5.5 V		100	180	ns
Pulse width of programming voltage	t_P	Chip disabled, V_{CC} = 5.5 V	1		40	µs
Required current limit of power supply feeding programming pin and output during programming	I_L	V_{PP} = 33 V, V_{OUT} = 26 V, V_{CC} = 5.5 V	240			mA
Required time delay between disabling memory output and application of output programming pulse	T_{D1}	Measured at 10% levels	70	80	90	µs
Required time delay between removal of programming pulse and enabling memory output	T_{D2}	Measured at 10% levels	100			ns
Output current during verification	I_{OLV1}	Chip enabled, V_{CC} = 4.2 V	11	12	13	mA
	I_{OLV2}	Chip enabled, V_{CC} = 6.2 V	0.19	0.2	0.21	mA
Address input voltage	V_{IH}		2.4	5.0	5.5	V
	V_{IL}		0.0	0.4	0.8	V
Maximum duty cycle during automatic programming of programming pin and output pin	D.C.	t_P/t_C			25	%

1/ T_C = +25°C.

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4.13 Programming procedures for circuit G. The programming characteristics of table IVG and the following procedures shall be used for programming:

a. Connect the device in the electrical configuration for programming. The waveforms on figure 6G and the programming characteristics of table IVG shall apply to these procedures.

b. Select the desired word by applying high or low levels to the appropriate address inputs. Disable the device by applying a high level to one or more 'active low' chip enable inputs. Note that address and enable inputs must be driven with TTL logic levels during programming and verification.

c. Increase V_{CC} from nominal to V_{CCP} (10.5 ±0.5 V) with a slew rate limit of I_{RR} (1.0 to 10.0 V/µs). Since V_{CC} is the source of the current required to program the fuse as well as the I_{CC} for the device at the programming voltage, it must be capable of supplying 750 mA at 11.0 volts.

d. Select the output where a logical high is desired by raising that output voltage to V_{OP} (10.5 ±0.5 V). Limit the slew rate to I_{RR} (1.0 to 10.0 V/µs). This voltage change may occur simultaneously with the V_{CC} increase to V_{CCP}, but must not precede it. It is critical that only one output at a time be programmed since the internal circuits can only supply programming current to one bit at a time. Outputs not being programmed must be left open or connected to a high impedance source of 20 kΩ minimum (remember that the outputs of the device are disabled at this time).

e. Enable the device by taking the chip enables to low level. This is done with a pulse PWE for 10 µs. The 10 µs duration refers to the time that the circuit (device) is enabled. Normal input levels are used and rise and fall times are not critical.

f. Verify that the bit has been programmed by first removing the programming voltage from the output and then reducing V_{CC} to 5.0 V (±0.25 V). The device must be enabled to sense the state of the outputs. During verification, the loading of the output must be within specified I_{OL} and I_{OH} limits.

g. If the device is not to be tested for V_{OH} over the entire operating range subsequent to programming, the verification of step f is to be performed at a V_{CC} level of 4.0 volts (±0.2 V). V_{OH}, during the 4-volt verification, must be at least 2.0 volts. The 4-volt V_{CC} verification assures minimum V_{OH} levels over the entire operating range.

h. Repeat 4.13b through 4.13f for each bit to be programmed to a high level. If the procedure is performed on an automatic programmer, the duty cycle of V_{CC} at the programming voltage must be limited to a maximum of 25 percent. This is necessary to minimize device junction temperature. After all selected bits are programmed, the entire contents of the memory should be verified.

i. For classes S and B devices, if any bit does not verify as programmed it shall be considered a programming reject.

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TABLE IVG. Programming characteristics for circuit G.

Characteristic	Symbol	Conditions	Limits 1/			Unit
			Min	Recommended	Max	
Required V_{CC} for programming	V_{CCP}		10.0	10.5	11.0	V
I_{CC} during programming	I_{CCP}	V_{CC} = 11 V			750	mA
Required output voltage for programming	V_{OP}		10.0	10.5	11.0	V
Output current while programming	I_{OP}	V_{OUT} = 11 V			20	mA
Rate of voltage change of V_{CC} or output	I_{RR}		1.0		10.0	V/µs
Programming pulse width (enabled)	P_{WE}		9	10	11	µs
Required V_{CC} for verification	V_{CCV}		3.8	4.0	4.2	V
Maximum duty cycle for V_{CC} at V_{CCP}	M_{DC}			25	25	%
Address setup time	t_1		100			ns
V_{CCP} setup time	t_2	2/	5			µs
V_{CCP} hold time	t_5		100			ns
V_{OP} setup time	t_3		100			ns
V_{OP} hold time	t_4		100			ns

1/ T_C = +25°C.

2/ V_{CCP} setup time may be greater than 0 if V_{CCP} rises at the same rate or faster than V_{OP}.

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5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-38510.

6. NOTES

6.1 Notes. The notes specified in MIL-M-38510 are applicable to this specification.

6.2 Intended use. Microcircuits conforming to this specification are intended for original equipment design applications and logistic support of existing equipment.

6.3 Ordering data. The acquisition document should specify the following:

a. Complete part number (see 1.2).

b. Requirements for delivery of one copy of the quality conformance inspection data pertinent to the device inspection lot to be supplied with each shipment by the device manufacturer, if applicable.

c. Requirements for certificate of compliance, if applicable.

d. Requirements for notification of change of product or process to the contracting activity in addition to notification to the qualifying activity, if applicable.

e. Requirements for failure analysis (including required test condition of method 5003 of MIL-STD-883), corrective action, and reporting of results, if applicable.

f. Requirements for product assurance options.

g. Requirements for special lead lengths or lead forming, if applicable. Unless otherwise specified, these requirements will not apply to direct purchase by or direct shipment to the Government.

h. Requirements for programming the device, including processing option.

i. Requirements for "JAN" marking.

6.4 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-M-38510, MIL-STD-1331, and as follows:

GND - - - - - - - - - - - - - - Electrical ground (common terminal).

I_{IN} - - - - - - - - - - - - - Current flowing into an input terminal.

V_{IC} - - - - - - - - - - - - - Input clamp voltage.

V_{IN} - - - - - - - - - - - - - Voltage level at an input terminal.

6.5 Logistic support. Lead materials and finishes (see 3.3) are interchangeable. Unless otherwise specified, microcircuits acquired for Government logistic support will be acquired to device class B (see 1.2.2), and lead finish C (see 3.3). Longer length leads and lead forming shall not affect the part number. It is intended that spare devices for logistic support be acquired in the unprogrammed condition (see 3.7.1) and programmed by the maintenance activity, except where use of quantities for devices with a specific program or pattern justify stocking of preprogrammed devices.

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6.6 Substitutability. The cross-reference information below is presented for the convenience of users. Microcircuits covered by this specification will functionally replace the listed generic-industry type. Generic-industry microcircuit types may not have equivalent operational performance characteristics across military temperature ranges or reliability factors equivalent to MIL-M-38510 device types and may have slight physical variations in relation to case size. The presence of this information shall not be deemed as permitting substitution of generic-industry types for MIL-M-38510 types or as a waiver of any of the provisions of MIL-M-38510.

Military device type	Generic-industry type/manufacturer	Circuit designator	Fusible link	Symbol/ CAGE number
01	7684/Harris Corporation	A	NiCr	CDWD/34371
01	77S184/National Semiconductor	G	TiW	CCXP/27014
01	82S184/Signetics Corporation	C	NiCr	CDKB/18324
02	7685/Harris Corporation	A	NiCr	---
02	77S185/National Semiconductor	G	TiW/W	---
02,10	82S185A/Signetics Corporation	C	NiCr	---
02,08	29651/Raytheon Company	F	NiCr	CRP/07933
03	77S180/National Semiconductor	G	TiW	---
03	7680/Harris Corporation	A	NiCr	---
03	82S180/Signetics Corporation	C	NiCr	---
03	93Z450/Fairchild Corporation	D	ZVE*	CFJ/07263
03	27S180/Advanced Micro Devices, Inc.	E	Platinum silicide	CDWN/34335
04	77S181/National Semiconductor	G	TiW/W	---
04	7681/Harris Corporation	A	NiCr	---
04,09	82S181A/Signetics Corporation	C	NiCr	---
04,09	93Z451/Fairchild Corporation	D	ZVE*	---
04	27S181/Advanced Micro Devices, Inc.	E	Platinum silicide	---
04	29631/Raytheon Company	F	NiCr	---
05	82S2708/Signetics Corporation	C	NiCr	---
05	93Z461/Fairchild Corporation	D	ZVE*	---
06	93Z460/Fairchild Corporation	D	ZVE*	---
02,08	53S841/Monolithic Memories, Inc.	B	TiW	---

* Zapped vertical emitter.

6.7 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Army - ER
Navy - EC
Air Force - 17

Review activities:
Army - AR, MI
Navy - OS, SH, TD
Air Force - 11, 19, 85, 99
DLA - ES

User activities:
Army - SM
Navy - AS, CG, MC

Preparing activity:
Air Force - 17

Agent:
DLA - ES

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